

On the Interplay of Explainability and Fairness in AI: A Survey

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(Survey Paper)

Abstract—Algorithmic fairness and explainability are foundational pillars of responsible AI. Although often studied independently, their interplay is increasingly recognized as crucial for diagnosing and mitigating bias in machine learning systems. We first introduce two systematic taxonomies: one for algorithmic fairness and one for explainable AI, to organize the landscape of existing work across diverse tasks (classification, ranking, and recommendation) and data modalities (tabular, graph). Next, we categorize the use of explanations in fairness efforts into three main functions: (a) detecting and understanding the causes of unfairness, (b) defining enhanced fairness metrics, and (c) designing mitigation strategies. In addition, we examine how explanation methods themselves can be biased, underscoring the need to evaluate fairness for explanations. Finally, we identify open research challenges and outline promising directions for future research at the intersection of fairness and explainability.

Index Terms—Fairness, explainability, responsible AI.

I. INTRODUCTION

THE rapid integration of artificial intelligence (AI) into high-stakes decision-making has intensified concerns about fairness and transparency. Machine learning models may reinforce existing biases or lead to discriminatory outcomes [1], while their increasing complexity and opacity make it difficult to understand, evaluate, and contest their decisions. To address these concerns, algorithmic fairness (ensuring unbiased outcomes) and explainability (interpreting model decisions) have become key dimensions of responsible AI [2], particularly in critical domains such as healthcare, law, and finance, where decisions can have profound impacts [3]. Once limited to academic research, these concerns are now reflected in emerging governance frameworks (e.g., the EU AI Act) that mandate non-discrimination and transparency in automated decision-making, emphasizing the growing importance of ensuring that machine learning models are both fair and explainable.

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Despite their shared importance in responsible AI, the interplay between fairness [3], [4] and explainability [5], [6] is still poorly mapped. Typically, these two objectives are examined independently: researchers developing fairness-aware models have not always prioritized interpretability, and conversely, work on explainable AI (XAI) often overlooks issues of bias. Therefore, existing literature surveys reflect the historical separation of these fields. However, there is an increasing realization that fairness and explainability are deeply intertwined rather than orthogonal goals. Detecting bias often requires meaningful explanations of model behavior, while fairness considerations can also shape what counts as a “good” explanation. For example, an AI system may provide detailed explanations to one demographic group but vague or incomplete explanations to another. Although recent studies have begun to examine these interactions, the precise connections between XAI and fairness remain insufficiently understood. This gap motivates our survey, which examines the interplay between explainability and algorithmic fairness in machine learning.

Related work: The work in [7] provides an early, high-level state-of-the-art overview of the relationship between model explanations and AI fairness, emphasizing how different explanation types influence human perceptions of fairness. While this work provides useful insights into how explanations influence fairness judgments made by users, its scope is relatively narrow, focusing mainly on human-centered evaluation. In contrast, our survey adopts a broader perspective: we introduce dual taxonomies of fairness and explainability and review technical approaches that connect the two across diverse tasks. Similarly, the work in [8] narrows the discussion to the natural language processing (NLP) domain, reviewing how existing XAI methods are used to detect and mitigate bias in language models and identifying barriers to using explainability for fair NLP systems. Our survey, by contrast, integrates explainability–fairness research across a range of machine learning tasks and data modalities.

The work in [1] adopts a meta-analytical perspective, critically examining common claims about the fairness benefits of XAI. It identifies seven recurring assumptions and shows that they are often vague or misaligned with the capabilities of current explainability methods. While this perspective clarifies the limits of XAI for fairness, it remains high-level. In contrast, our survey organizes concrete explanation-based fairness techniques into detailed taxonomies, with examples across domains. More recently, the work in [9] introduced a conceptual

framework for FairXAI, proposing a high-level taxonomy and the FairXAI wheel, which outlines four desirable properties: quality, robustness, fairness, and sparsity for trustworthy explanations. While their framework serves as a valuable conceptual guide, the discussion remains largely abstract and lacks detailed algorithmic and task-specific analysis. In contrast, our survey provides a more granular and technically grounded analysis. Specifically, our survey introduces comprehensive taxonomies for fairness and explanation methods, explicitly categorizes how explanations contribute to fairness, and systematically reviews concrete methods across multiple domains and tasks. It also examines the fairness of explanations themselves in greater depth.

Scope and contributions: This survey explores the intersection of algorithmic fairness and explainability from a technical perspective. Focusing on recent methodological advances, it delivers a comprehensive, integrative overview of explainable fairness and a detailed analysis of specific techniques and their interactions. Our central use case is classification on tabular data, one of the most common settings in fairness research; however, to ensure broader applicability, we also cover recommendations, rankings, and graphs. We restrict our scope to works that use GNNs strictly as black-box predictors, without addressing broader learning tasks.

First, we introduce systematic taxonomies that organize the landscape of approaches of fairness and explainability, and a comprehensive analysis at the intersection of fairness and explainability, offering a unified framework that was absent in earlier surveys. Second, we categorize existing explanation methods by their role in supporting fairness and survey representative methods in each category. In particular, we distinguish between explanation techniques that 1) detect bias and/or understand the causes of (un)fairness, 2) enhance fairness metrics, and 3) mitigate unfairness. Third, we extend the discussion to fair explanations by highlighting how explanation methods themselves can be subject to bias. Finally, we identify open challenges and research gaps that emerge at the intersection of fairness and explainability. A preliminary version of this paper appeared in [2].

Paper structure: Section II provides the necessary preliminary definitions and introduces our taxonomies of fairness and explanation approaches. Section III delves into the core of explainable fairness: it explores how explanations can be used to assess and improve fairness, structuring the discussion around the three major categories of methods. Section IV reviews work on fairness of explanations. Section V concludes the paper with a summary of our key findings and a reflection on the path forward for research in explainable fairness.

II. PRELIMINARIES

Fairness notions and explainability approaches depend on both the task type (classification, ranking, recommendation) and the data modality (tabular, graph).

Classification considers an input space $X = (\mathbf{x}_1, \dots, \mathbf{x}_N)$ of N points and an output space Y with ground-truth labels y_i ; a function $f : X \rightarrow Y$ learns the mapping, and $f(\mathbf{x}_i) = \hat{y}_i$ denotes the predicted label for an unseen input while y_i

is the ground-truth label. In ranking, given an item set $I = (i_1, \dots, i_M)$, a utility function $u : I \rightarrow \mathbb{R}$ assigns a score to each item, and sorting these scores yields a permutation $\pi : I \rightarrow \{1, \dots, M\}$ such that $u(i_a) > u(i_b) \Rightarrow \pi(i_a) < \pi(i_b)$, i.e., higher-utility items appear earlier in the ranked list. Recommendation considers a user set U and an item set I ; a relevance function $\hat{s} : U \times I \rightarrow \mathbb{R}$ predicts user-item preference, and for each user u the system outputs $\mathcal{R}_u = \text{TopK}_{i \in I} \hat{s}(u, i)$, the k items with the highest predicted scores.

Turning to data modality, we first consider tabular data. A tabular dataset is a structured collection of instances, represented as a matrix $T \in \mathbb{R}^{N \times d}$, where N is the number of instances and d the number of features. Each instance $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector. On this representation a classifier learns a mapping $f : X \rightarrow Y$; a ranker learns a utility function $u : X \rightarrow \mathbb{R}$ whose sorted values yield a permutation of the instances; and a recommender learns a relevance function $\hat{s} : U \times X \rightarrow \mathbb{R}$.

Finally, in graph data, a graph is defined as $G = (V, E)$ with node set V and edge set $E \subseteq V \times V$; each node $v \in V$ has a feature vector $\mathbf{x}_v$, and edges may carry attributes. A classifier learns $f : V \rightarrow Y$ (or $f : E \rightarrow Y$) to label nodes or edges; a ranker learns a utility function $u : V \rightarrow \mathbb{R}$ (resp. $u : E \rightarrow \mathbb{R}$) whose scores yield an ordering of nodes (resp. edges); and a recommender learns $\hat{\ell} : V \times V \rightarrow \mathbb{R}$ to score potential links, returning the top- k highest-scoring pairs.

A. Fairness

This section first reviews core fairness notions, criteria, and stage strategies of the learning pipeline. While fairness is a broad term, most formalizations of algorithmic fairness focus on non-discrimination, formulated as the requirement that the output of an algorithm should not be influenced by attributes that are not relevant to the task at hand [3], [4], [10]. Such attributes are called *protected*, or *sensitive*, and often include gender, religion, age, and race.

Since there has been a proliferation of approaches to modeling, measuring, and addressing fairness, we present a taxonomy to help us place work in explaining unfairness into perspective (see Fig. 1).

Level of fairness: Fairness models are broadly distinguished as individual or group-based ones. *Individual* fairness asks that similar individuals are treated similarly. *Group* fairness assumes that individuals are partitioned into groups based on the value of one or more of their protected attributes and asks that the groups be treated similarly. We present briefly common formulations of individual and group fairness for classification.

Most individual-based fairness models are distance-based, where given a distance measure between individuals, the difference of the outputs of the classifier for two individuals is bound by the distance between them [11]. Alternatively, counterfactual fairness [12], [13] posits that an output is fair if it is consistent across both the actual world and a hypothetical scenario where the individual belongs to a different group.

Group fairness can be categorized into *distributive* group fairness, which focuses on the fairness of model outcomes, and *procedural (or process)* fairness, which evaluates the fairness of

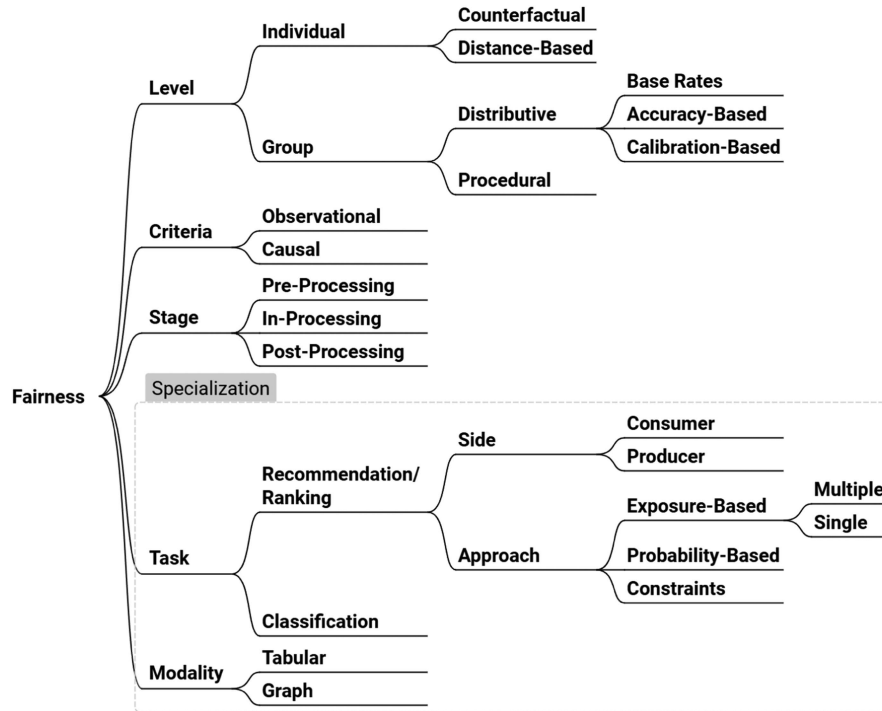


Fig. 1. Taxonomy of fairness approaches.

the decision-making process itself rather than just its results [14], [15].

For distributive group fairness, let us assume that there are two groups, the protected group G^+ and the non-protected group G^- . Let $f: X \rightarrow \{0, 1\}$ be a binary classifier with 1 being the favorable outcome, and $y, \hat{y}$ be the ground truth and the predicted output, respectively. Distributive group fairness can be further categorized into base-rate fairness, accuracy-based fairness and calibration-based fairness.

Base rate group fairness compares the probability $P(\hat{y} = 1 | v \in G^+)$ that an individual v receives a favorable outcome when v belongs to the protected group with the corresponding probability $P(\hat{y} = 1 | v \in G^-)$ when v belongs to the non-protected group. The most commonly used such fairness metric is *demographic parity*, which asks that these two probabilities are equal. *Accuracy-based* fairness warrants that various types of classification errors are equal across groups. For example, *equal opportunity* expressed as: $P(\hat{y} = 1 | y = 1, v \in G^+) = P(\hat{y} = 1 | y = 1, v \in G^-)$ considers true positive rate (TPR), while *equalized odds* requires the same TPR and false positive rate (FPR). Finally, *calibration-based* methods aim at equal calibration across groups in the case of probabilistic classifiers. Let S be the predicted probability. The goal is to ensure that $P(y = 1 | S = p, v \in G^+) = P(y = 1 | S = p, v \in G^-)$, where $p \in [0, 1]$.

Fairness criteria: Fairness definitions can also be approached based on two criteria [16]: *observational criteria*, relying solely on the observational distribution of data, and *causal criteria*. Observational criteria have a key limitation: they risk “correcting” disparities that may arise from genuine differences rather than unfair biases. Since these criteria rely solely on statistical patterns in the data, they do not distinguish between

disparities due to structural discrimination and those resulting from non-discriminatory differences in underlying characteristics. As a result, enforcing fairness based on observational metrics alone may lead to overcorrection, altering outcome distributions in a way that eliminates meaningful, contextually justified variations. To tackle this limitation, causal criteria go beyond observational patterns and seek to understand the underlying causal relationships between variables. One such approach is *counterfactual fairness* [12], which requires that an individual would receive the same outcome in a counterfactual world where their sensitive attribute (e.g., race or gender) is different, while all other relevant characteristics remain unchanged.

Stage of fairness: Methods for mitigating unfairness can be categorized into three distinct groups based on the *stage* at which they are applied [3], [4]: (a) *Pre-processing* approaches modify the input data, (b) *In-processing* approaches modify the model so that it produces fair results for any input, and (c) *Post-processing* approaches modify the output.

Fairness for ranking and recommendation: Beyond classification, bias in ranking and recommender outputs manifests as unequal exposure or utility across user- or item-side groups. A common taxonomy distinguishes individual versus group and consumer versus producer perspectives and employs metrics such as exposure parity, long-tail (popularity-bias) gap, and consumer-provider balance [10]. Static parity metrics, however, miss systemic effects—including utility gaps, temporal spill-overs, strategic behaviour, and statistical uncertainty, so stakeholder-centred, impact-oriented evaluation is required [17]. In score-based and learning-to-rank settings, fairness notions (exposure and relevance parity, calibration) align with three families of mitigation: re-ranking, constrained optimisation,

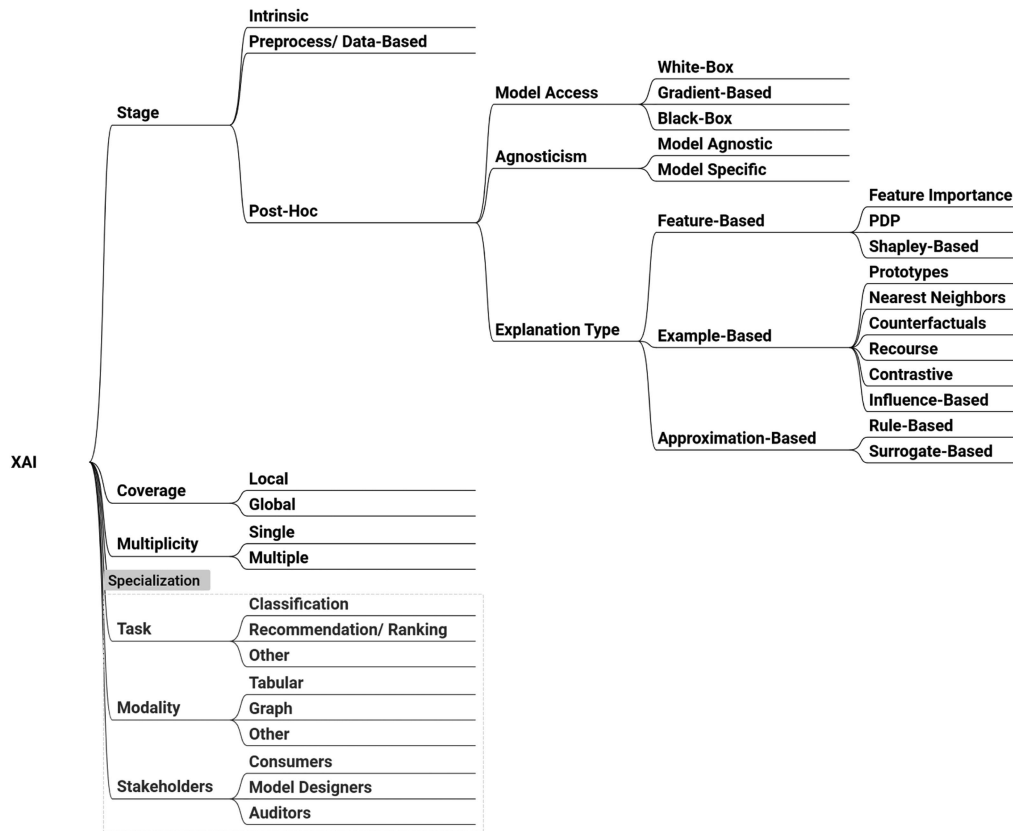


Fig. 2. Taxonomy of explanation approaches.

and stochastic exposure policies [18], [19]. Recommender systems assess fairness at both user and item levels. Mitigation methods include adversarial debiasing and multi-objective training [20]. Because bias removed in pre-processing can re-emerge later, constraints should span pre-, in-, and post-processing stages [21]. Current directions explore cross-domain fairness transfer [22] and the fairness–diversity interplay, showing that exposure-based corrections may help or harm recommendation diversity and motivating joint fairness–diversity optimisation [23].

Fairness for graphs: Unlike tabular data, graph-structured settings introduce statistical dependence, topology-induced bias, homophily-driven amplification, and an interplay between edge- and node-level fairness that complicate standard mitigation approaches. Therefore, classical notions like demographic parity, equal opportunity, individual, and counterfactual fairness are re-interpreted over graph primitives: demographic parity, equal opportunity, and individual fairness are lifted from nodes to links and subgraphs by averaging predictions over edge sets or node subsets, while counterfactual fairness is adapted by perturbing sensitive attributes together with the edges they influence, and degree fairness enforces parity with respect to node degree [24]. Bias mitigation aligns with the standard ML pipeline: pre-processing (edge rewiring, feature masking, synthetic debiasing graphs), in-processing (statistical-parity regularizers, adversarial debiasing), and post-processing (threshold shifts, fair re-ranking, exposure balancing) [25], [26]. However, structural interventions, such as rewiring alone, cannot eliminate

visibility bias, the tendency of well-connected nodes to gain more exposure [27]. Moreover, the non-IID (not independent and identically distributed) nature of graphs means perturbing a single attribute can cascade through neighbors, making graph-aware counterfactual evaluation essential [28].

B. Explanations

To assist us in understanding the connection between fairness and explainability, in Fig. 2, we present a taxonomy that provides a general overview of the broad field of explainable AI [5], [6], [29], [30], [31], [32], [33].

Stage: Explanation methods can appear at different stages in the machine learning pipeline as: (a) intrinsic, (b) pre-process or data-based, and (c) post-hoc ones.

Intrinsic methods are crafted purposefully for transparent and interpretable outcomes [30], [32]. These models are known for their simplicity and inherent transparency, allowing users to understand the reasoning behind each output. However, they may sacrifice accuracy when dealing with complex data patterns [31], [32].

Preprocess or data-based approaches utilize unsupervised techniques to uncover hidden data patterns and structures, aiding in enhancing the understanding of the dataset prior to model training. Techniques such as principal component analysis (PCA) [34], singular value decomposition (SVD) [35], non-negative matrix factorization (NMF) [36], independent component analysis (ICA) [37], factor analysis (FA) [38], and

t-distributed stochastic neighbor embedding (t-SNE) [39] are commonly employed for dimensionality reduction, feature extraction, and clustering.

Post-hoc approaches are applied after model training to provide explanations for complex machine learning models lacking inherent interpretability. Post-hoc explanations are the ones most commonly used in explaining unfairness. We further structure them by (a) model access, (b) model agnosticism, and (c) explanation type.

Model access: The level of access to the model internals significantly impacts how explanations are generated. It can be broadly categorized into: (i) complete model access/white-box, (ii) access to gradient and (iii) black-box. *Complete model access/white-box* approaches offer the algorithm unfettered access to the intricate internal mechanics of the machine learning model [29]. This granularity allows for the creation of precise and context-aware explanations, optimizing the potential for achieving high fidelity to the internal operations of the model. Explanation methods that utilize *access to gradients*, often through optimizers like Adam [40], are effective when gradient information is accessible. This approach offers a balance between model specificity and versatility, enabling informative explanations without full model access. Finally, when the algorithm operates without direct access to the internal details of the model, it falls under the category of a *black-box approach* [6], [29], [30].

Model agnosticism: Model agnosticism refers to the capability of the algorithm to operate on a wide range of model architectures without being bound by specific model requirements. It characterizes the adaptability of an explanation method across different types of machine learning models [6], [29], [30], [31], [32], [33], [41]. While most black-box approaches are model agnostic, black-box models that require specific input features or domain knowledge are not. Conversely, *model-specific* approaches can be used to interpret only a specific type of black-box model, constraining their applicability and versatility to a narrower spectrum of machine learning models. Most of the model-specific approaches have gradient or white-box access to the model.

Explanation type: Explanations can be characterized as: feature-based, example-based, and approximation-based.

Feature-based explanations [6], [31] assess the significance of individual features in influencing the prediction of a model. Feature-based methods, including *feature importance* [42], *partial dependence plots (PDPs)* [43], and *Shapley additive explanations (SHAP)* [44], assign scores to input features, indicating their relative importance when a prediction is made.

Example-based approaches [29], [31], [32], [33], provide explanations through specific instances in the vicinity of the explained data point [29]. They encompass several techniques such as *counterfactual explanations* [45], determining what should be the change in features of an individual to cross the decision boundary of a classifier, and *recourse based* [46] explanations providing actionable changes for the same purpose [47]. Another approach is using *prototypes* that utilize a set of representative instances from a class [31], *nearest neighbors* that leverage neighboring data points [6], and *influence-based* methods which

trace predictions back to specific training instances that significantly influenced a decision [48], [49]. Additionally, *contrastive explanations* [47], [50], contrasting the presence and absence of specific features [31].

Finally, *approximation-based* methods [29] aim to approximate the behavior of the underlying model and the decision boundaries in the vicinity of the explained datapoint, using interpretable models or simpler proxies. Approaches within this category include *surrogate based* [5], [51] explanations, which fit linear models near the explained datapoint, and *rule based* ones [51], which extract rules from these models [29].

Coverage: We can distinguish explanations in two key types: global and local explanations [29], [30], [31], [32], [33]. *Global explanations* aim to elucidate the overall logic and decision-making process of a machine learning model. *Local explanations* focus on individual predictions, offering specific insights into the reasons behind a particular decision.

Multiplicity: An explanation method may produce a single or multiple explanations. Single explanations are concise and easy to understand, but may not capture the full complexity of the model. Multiple explanations provide a more detailed understanding, but can be harder to interpret [29].

Explanations for ranking and recommendations: Moving from classification to ranking and recommendation, explanations span three core categories: *feature-based*, *example-based*, and *approximation-based* [52], [53]. These categories span models from traditional tabular approaches to recent graph-based recommenders. Feature-based explanations pointing to item or user attributes, such as content features, keywords, or latent factors mapped to interpretable traits, that align with the user's known preferences [53]. Example-based explanations cite specific instances, such as similar users or liked items (for example, "this item is recommended because you liked X" or "users with similar tastes enjoyed Y") [54], [55]. *Approximation-based* explanations learn an interpretable surrogate (e.g., a compact decision tree or a sparse feature-weighting model) or extract human-readable rules to mimic the behaviour of the original, complex recommender, thereby exposing its decision logic while aiming to preserve output fidelity [53], [56].

Explanations for graphs: Shifting from tabular data to graphs, explanation techniques have been developed for both classical graph-based models and Graph Neural Networks (GNNs). Methods are commonly grouped into three families: node- or edge-level attribution, surrogates, and counterfactual subgraphs [57]. At a high level, graph explainers identify which nodes, edges, or subgraphs together with their attributes or edge types, influence the prediction of a model [58]. Standard evaluation relies on fidelity, sparsity, and stability [57], supplemented by a perturbation-resilience score that quantifies robustness to small graph edits [59]. Comparative benchmarks reveal that gradient-based attribution achieves higher fidelity scores on homophilous graphs, whereas perturbation-based surrogates are more reliable under heterophily [60]. Counterfactual explainers generate minimal-edit counterfactual subgraphs and can be evaluated using graph-specific benchmarks such as the GRETEL framework [61]. A broad cross-model study shows explanation fidelity can degrade sharply under distribution shift, motivating

sanity checks, e.g., integrated-gradients explanations verified by edge-level perturbations [30]. Recent taxonomies distinguish post-hoc explainers from self-interpretable GNNs and emphasize scalability, fairness, and privacy as open challenges [62]. Knowledge graphs add a complementary, logic-based layer: they supply semantic paths, human-readable rules, and ontology context for explanations, with contributions organized by stage (*pre-, in-, post-model*) and function (*feature engineering, relation extraction, KG construction, reasoning*) [58], [63].

Stakeholder-specific explanations: These should be tailored to the stakeholders they serve, since each group differs in goals, expertise, and expectations. Prior work characterizes stakeholders by expertise levels [64], [65], [66] or functional roles such as model builders, breakers, end users, or examiners [67], [68], [69]. However, expertise-based scales oversimplify needs and role-based ones lack composability, which [70] addresses by decomposing stakeholders into knowledge, context, goals, and tasks. We propose three groups, each tied to a different explanation type: (a) *Consumers*, the recipients of model decisions, need explanations to guide or contest outcomes and benefit most from example-based explanations—counterfactuals, recourse, contrastive, or nearest neighbors—that show how small changes alter outcomes; (b) *Model designers*, who build and tune the model, use feature-based explanations (feature importance, *PDPs*, *Shapley-based*) and approximation-based ones (surrogate, rule-based) to see how features drive predictions and locate bias or instability; (c) *Auditors*, who assess robustness, transparency, and compliance, rely on approximation- or rule-based explanations with global scope that summarize behavior interpretably and prefer model-agnostic methods that generalize and support repeatable evaluation.

III. EXPLANATIONS FOR FAIRNESS

In this section, we explore how we can use explanations to address unfairness. We identify the following three directions:

- *Explanation methods used to detect biases and/or understand the causes of (un)fairness:* By explaining the behavior of a model, such methods aim at discovering unfair behavior, for example, by detecting dependencies of the output on protected features. Explanations are also used to identify potential causes of unfairness, for example, by identifying the features that contribute to unfavorable outcomes for specific groups.
- *Explanation methods used for defining enhanced fairness metrics:* These methods focus on proposing new metrics to quantify (un)fairness, most often by measuring the cost of mitigation actions for achieving fairness.
- *Explanation methods used for mitigating unfairness:* These methods propose new approaches to alleviate unfairness, for example, by reducing dependencies on sensitive attributes or by recommending specific actions to counteract unfairness.

Given the various forms of unfairness that different approaches are addressing, one distinctive characteristic is the type of unfairness they are considering. First, we can distinguish the different approaches based on the level of fairness, that is,

whether it is at the individual or group level. At the group level, one common approach is to generate explanations for individuals and then aggregate them. Several aggregation techniques have been proposed, including SP-LIME [71], which selects a representative subset of instances as explanations by prioritizing features with high global importance. This global importance is typically measured as the square root of the sum of absolute local attributions. The core assumption is that features that frequently appear with high importance in local explanations also reflect the global behavior of the model. However, recent studies caution that while local explanations are reliable for individual predictions, they may be misleading when generalized to group or global model behavior [72]. An alternative approach is to define group-level explanations, where similar instances are explained collectively. These methods often group instances by protected attributes (e.g., race, gender) and generate explanations for these groups. Unlike individual explanations, group explanations analyze model behavior across entire demographic groups and can reveal systemic biases and disparities. While group explanations can be useful for fairness audits, they may lack accuracy for individual cases and overlook subgroup variations.

Then, we can further differentiate the different approaches based on the specific notion of fairness they address. For instance, some methods focus on distributional fairness, such as demographic parity, equalized opportunity, or equalized odds. Methods focusing on demographic parity typically generate explanations for all positive instances, as this fairness criterion requires that the rate of positive outcomes be equal across different demographic groups. Similarly, for equalized opportunity, they generate explanations for true positives and for equalized odds for both true and false positives. Other approaches focus on procedural fairness, which evaluates how decisions are made rather than just what outcomes are produced. These methods typically analyze the attributions assigned to input features across demographic groups, investigating whether protected attributes (or their proxies) are disproportionately used for certain groups.

Another consideration is the treatment of the protected attribute in the explanation process. On one hand, including protected attributes can be essential for identifying cases of disparate treatment or direct discrimination, as it allows to observe how model decisions vary across sensitive groups. On the other hand, excluding protected attributes may help uncover indirect discrimination or disparate impact by revealing the reliance of the model on proxy features that are highly correlated with the protected attribute.

Most of the current research utilizes counterfactual explanations, while a few are feature-based and use Shapley values. One study [73] proposes a flexible pipeline, implemented in MIMOSA [74], for comparing group fairness metrics with explanation-based findings, identifying features contributing to biased outcomes, selecting explanation methods through evaluation metrics, visualizing explanations from multiple perspectives and tracking how fairness interventions affect feature contributions. Tables I and II summarize the classification and recommendation/ranking methods reviewed in this section, respectively, with respect to our fairness and explanation dimensions.

TABLE I

OVERVIEW OF APPROACHES FOR EXPLAINING (UN)FAIRNESS IN CLASSIFICATION. **MODALITY (MOD.):** TAB = TABULAR, GR = GRAPH. **ACCESS (ACC.):** ● BLACK-BOX, ① GRADIENT ACCESS, ○ WHITE-BOX. **AGNOSTIC (AGN.):** ✓ MODEL AGNOSTIC, ✗ MODEL SPECIFIC. **COVERAGE (COV.):** ① GLOBAL, ② LOCAL. **LEVEL:** ① INDIVIDUAL, ② GROUP. **CRITERIA:** OBS. = OBSERVATIONAL, CAUS. = CAUSAL. **GOAL:** ① ENHANCE METRICS, ② UNDERSTAND CAUSES, ③ MITIGATE UNFAIRNESS.

Appr.	Mod.	Explanation					Fairness			
		Stage	Acc.	Agn.	Cov.	Type	Level	Type	Criteria	Goal
[90]	Tab	Post	●	✓	②	CFEs: Most significant feature change	②	Implicit/Explicit bias	Obs.	②
[87]	Tab	Post	●	✓	②	CFEs	① ②	Burden	Obs.	① ②
[89]	Tab	Post	●	✓	①	CFEs: Burden	① ②	Burden	Obs.	① ②
[94]	Tab	Post	●	✓	①	CFEs	②	Fairness of recourse	Obs.	②
[96]	Tab	Post	●	✓	①	CFEs: Subgroup recourse actions	②	Fairness of recourse	Obs.	① ②
[103]	Gr	Post	●	✗	①	CFEs: Per-edge importance score	②	Base-rates	Obs.	② ③
[104]	Gr	Post	①	✗	① ②	CFEs + Recourse: Diagnostic + actionable subgraph	②	Base-rates	Obs.	② ③
[93]	Tab	Post	●	✓	②	Recourse: Two level recourse sets	① ②	Procedural	Obs.	②
[84]	Tab	Post	●	✓	①	Recourse: Causal path	②	Base-rates	Caus.	② ③
[105]	Tab	Post	●	✓	①	Recourse	②	Fairness of recourse	Obs.	① ②
[106]	Tab	Post	●	✓	②	Recourse	① ②	Fairness of recourse	Caus.	① ②
[92]	Tab	Post	●	✓	②	Contrastive: Probabilistic contrastive CFEs	① ②	Fairness of recourse	Caus.	②
[98]	Gr	Post	①	✗	②	Contrastive: Edges (structural)	①	Base-rates	Obs.	① ② ③
[83]	Tab	Post	●	✓	②	Shapley-based: Visualization	②	Base-rates	Obs.	② ③
[79]	Tab	Post	●	✓	②	Shapley-based: Feature values	②	Base-rates/Accuracy-based	Obs.	②
[80]	Tab	Post	●	✓	②	Feature-based: Feature values	① ②	Procedural	Obs.	① ②
[75], [76]	Tab	Post	●	✓	①	Surrogate-based: Global important features	②	Procedural	Obs.	① ② ③
[85]	Tab	Post	●	✓	①	Data/Influence-based: Top-k data subsets	②	Base-rates/Accuracy-based	Obs.	② ③
[48]	Tab	Post	①	✗	①	Influence-based: Predicate-based causal	②	Base-rates/Accuracy-based	Obs.	② ③
[100]	Gr	Post	○	✗	①	Influence-based: Top-k influential training nodes	②	Base-rates	Obs.	② ③

TABLE II

OVERVIEW OF APPROACHES FOR EXPLAINING (UN)FAIRNESS IN RECOMMENDATION (REC) AND RANKING (RANK). **MODALITY (MOD.):** TAB = TABULAR, GR = GRAPH. **ACCESS (ACC.):** ● BLACK-BOX, ① GRADIENT ACCESS, ○ WHITE-BOX. **AGNOSTIC (AGN.):** ✓ MODEL AGNOSTIC, ✗ MODEL SPECIFIC. **COVERAGE (COV.):** ① GLOBAL, ② LOCAL. **LEVEL:** ① INDIVIDUAL, ② GROUP. **CRITERIA:** OBS. = OBSERVATIONAL, CAUS. = CAUSAL. **GOAL:** ① ENHANCE METRICS, ② UNDERSTAND CAUSES, ③ MITIGATE UNFAIRNESS.

Appr.	Mod.	Explanation					Fairness			
		Stage	Acc.	Agn.	Cov.	Type	Level	Type	Criteria	Task
[110]	Gr	Post	○	✗	②	CFEs: Removed user-item edges	① ②	Exposure-based + Consumer/Producer	Obs.	Rec
[108]	Gr	Post	●	✓	①	CFEs: Attribute-level changes	②	Exposure-based + Producer	Obs.	Rec
[107]	Tab	Post	①	✓	①	CFEs: Top-feature perturbations	②	Exposure-based + Producer	Obs.	Rec
[101]	Gr	Post	①	✗	①	CFEs: Counterfactual user-item edges	②	Constraints + Consumer	Obs.	Rec
[102]	Gr	Pre	○	✗	①	CFEs: Additional user-item edges	②	Constraints + Consumer	Obs.	Rec
[109]	Gr	Post	●	✓	① ②	CFEs: Per-user/item fairness contribution	① ②	Base-rates + Consumer/Producer	Obs.	Rec
[115]	Tab	In	○	✓	②	CFEs: Counterfactual ranking scores	②	Exposure-based + Producer	Caus.	Rec
[112]	Tab	Post	●	✓	①	Shapley-based: Attribute-level	②	Constraints + Producer	Obs.	Rank
[113]	Tab	Post	●	✓	①	Shapley-based: Attribute-level	②	Constraints + Producer	Obs.	Rank
[111]	Tab	Post	●	✓	①	Approximation-based: Regression on data characteristics	②	Constraints + Consumer/Producer	Obs.	Rec
[99]	Gr	Post	●	✓	②	Example-based: Top-k KG-path	① ②	Constraints + Consumer	Obs.	Rec

A. Explanations Using LIME

LIME (Local Interpretable Model-agnostic Explanations) builds an interpretable surrogate model that locally mimics the behavior of the black-box model [71]. Specifically, let f denote the function learned by a classification or a regression model. Given a specific instance $\mathbf{x}$, LIME constructs a model $g \in G$, where G represents a set of interpretable models, such as linear models or decision trees. This surrogate model g approximates

f within the local neighborhood of $\mathbf{x}$. LIME generates neighborhood instances around $\mathbf{x}$ and assigns a weight vector $\pi_{\mathbf{x}}(\mathbf{z})$ to each generated instance $\mathbf{z}$ based on its proximity to $\mathbf{x}$. Then, LIME optimizes

$$\arg \min_{g \in G} \mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g),$$

where $\mathcal{L}(f, g, \pi_{\mathbf{x}})$ measures the quality of the approximation of f by g in the locality of $\pi_{\mathbf{x}}$ and $\Omega(g)$ is a measure of the

complexity of g , for example, the number of non-zero weights, when g is a linear model.

LimeOut [75], [76] uses LIME to detect and mitigate unfairness. It first uses LIME to generate explanations for a selected set of instances. The instances to be explained are chosen using the Submodular Pick (SP) algorithm proposed in [71] that greedily selects instances to ensure that their explanations are diverse and not redundant.

LimeOut is grounded in the principles of procedural fairness. As opposed to looking at the fairness of the output, procedural fairness considers the degree to which individuals perceive the use of specific input attributes in decision-making as fair [14], [15]. LimeOut assumes that the use of protected attributes is inherently unfair. It characterizes a classifier as unfair if protected attributes are among the most important features contributing to the decision of the model.

When unfairness is detected, LimeOut seeks to increase the fairness of the classifier by reducing its dependency on protected attributes. To achieve this, LimeOut applies a feature drop. It generates a pool of classifiers where each classifier is trained with a subset of protected features removed. It then constructs an ensemble classifier through a linear combination of the classifiers in the pool to reduce the influence of protected attributes while preserving predictive performance.

Discussion: LIME approximates local model behavior with simple surrogate models, making feature reliance easy to inspect at a lower computational cost than more exhaustive attribution methods. However, LIME-based methods such as LimeOut may suffer from instability due to perturbation-based sampling and limited surrogate fidelity for complex models.

B. Explanations Using Shapley Values

Shapley values originate from cooperative game theory, where players form coalitions to achieve a common objective. The Shapley value assigns a fair distribution of the total payoff (or value) generated by the coalition among its members [77]. It ensures that the contribution of each player is fairly accounted for, based on both their individual impact and their interactions with others. In the context of classification, the Shapley value of a feature i in a set of base features F accounts for the contribution of i to a local prediction $f(\mathbf{x})$, denoted as the local Shapley value of i , or for the contribution of i to all predictions of the model, denoted as the global Shapley value of i . SHAP (SHapley Additive exPlanations) is a unified framework based on Shapley values that provides consistent and interpretable attributions of feature contributions by averaging their marginal impacts across all possible feature subsets [44]. For the local Shapley value of a feature i , and a feature coalition $S \subseteq F \setminus \{i\}$, a model $f_{S \cup \{i\}}$ is trained with feature i , while a model f_S is trained without it. The local Shapley value $\phi_{f(\mathbf{x})}(i)$ is the marginal contribution of i weighted and summed over all possible feature coalitions:

$$\phi_{f(\mathbf{x})}(i) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \cdot [f_{S \cup \{i\}}(\mathbf{x}_{S \cup \{i\}}) - f_S(\mathbf{x}_S)].$$

Various approaches have been proposed that estimate Shapley values without retraining the model [78]. The global Shapley value $\Phi_{f(X)}(i)$ of a feature i is computed by averaging the local Shapley values over all predictions:

$$\Phi_{f(X)}(i) = \mathbb{E}_{p(\mathbf{x}, y)}[\phi_{f(\mathbf{x})}(i)],$$

where $p(\mathbf{x}, y)$ is the joint distribution from which the labeled data is sampled.

Shapley value attribution is additive; the model output is explained as a sum of effects attributable to each of the input features. If this decomposition is exact, then it holds that:

$$f(\mathbf{x}) = \phi_0(f) + \sum_i \phi_{f(\mathbf{x})}(i),$$

where $\phi_0(f)$ is a constant bias term independent of $\mathbf{x}$.

For detecting unfairness, the authors of [79] leverage the SHAP values of the protected attribute, similarly to the LimeOut approach. The underlying assumption is again that the protected attributes should not influence the model decision. Specifically, it is assumed that demographic parity holds if the mean absolute SHAP value of the protected attribute is negligible. Other definitions of fairness are also considered. For satisfying equality of opportunity, it is tested whether the distributions of the SHAP value of the protected attribute of all demographic groups are similar for the instances of the positive class ($y = 1$). Finally, for equalized odds to hold, the assumption is that the distributions of the SHAP value of the protected attribute should be similar across groups both for instances in the positive class ($y = 1$) and for instances in the negative class ($y = 0$). A post-processing mitigation approach is also sketched. The main idea is to flip the decisions for selected members of the protected group, prioritizing those whose protected attribute has a high SHAP value.

Along similar lines, the authors of [80] propose a metric called GPF_{FAE} to evaluate group procedural fairness. The key idea is that similar individuals from different groups should receive similar explanations for their outcomes. To quantify this, the method constructs matched sets of similar individuals from different groups and compares the distributions of their explanations. Specifically, it measures the disparity between explanation distributions using Maximum Mean Discrepancy (MMD) with an exponential kernel, capturing differences in how the model processes inputs from each group. The authors of [81] relate SHAP-based feature importance to standard fairness measures by comparing feature contributions across protected and non-protected groups. They also assess whether reweighting-based debiasing changes SHAP importance in line with metrics such as disparate impact and equality of opportunity.

A more elaborate method in [82] uses Shapley values to measure not only the influence of the protected attribute, but also the influence of attributes that may serve as proxies. Consider decisions Y that are outcomes of a process acting on non-protected variables X and protected variables Z . This approach identifies *direct discrimination* generated by the influence of the protected attribute on the decision of a model, *indirect discrimination* that occurs when features X influenced by Z unfairly affect

Y , and *induced discrimination* that happens when removing direct discrimination inadvertently causes discrimination through features X associated with Z . For example, denying a loan just because of the race of the applicant (protected attribute Z) is direct discrimination. However, if certain neighborhoods used to decide loan approval (feature X) are predominantly one race, this may lead to indirect and potentially induced discrimination. The use of protected attributes or their proxy features is also linked to legal regulations. *Unfair influence* is an influence of protected features that is judged illegal via some legal instrument. *Fair relationship* of protected features with non-protected ones is a relationship that is judged legal via some legal instrument, e.g., a business necessity clause. To prevent discrimination, the authors propose to nullify the influence of the protected attribute on the model output while preserving the influence of the remaining features. To achieve this goal, a post-processing approach is proposed. The approach removes the influence of Z using the marginal direct effect (MDE) and introduces a loss function based on Shapley values to prevent increasing the impact of attributes X associated with Z (induced discrimination).

The authors of [83] employ Shapley values to attribute the influence of a feature on the fairness of the model as opposed to its influence on the output of the model. As a first step, the standard value function based on the model output $f(\mathbf{x})$ is replaced by a new value function that targets a chosen group-fairness metric. The resulting local Shapley values are then aggregated over the data into global Shapley values that attribute the model's overall (un)fairness to individual features. For demographic parity, this is captured by the disparity $D\Phi_f(i)$, i.e., the difference in the average Shapley contribution of feature i between the protected and non-protected groups:

$$D\Phi_f(i) = \mathbb{E}_{\mathbf{x} \in G^+} [\Phi_{f(X)}(i)] - \mathbb{E}_{\mathbf{x} \in G^-} [\Phi_{f(X)}(i)].$$

Due to the additive property of Shapley values, such explanations directly capture unfairness through their sum and are thus more robust to manipulation.

In addition, a meta-algorithm is proposed to achieve fairness. Specifically, given a model f , a parametric perturbation δ_θ is learned using existing in-processing algorithms for fairness, such that the resulting model $f_\theta = f + \delta_\theta$ is fair. By doing so, different perspectives are gained from the explanations of each component: the original sources of unfairness from f , the corrections made from δ_θ , and how the corrected model makes fair predictions from f_θ . This is possible because of the linearity axiom of the Shapley values that states that the fairness Shapley values of a linear ensemble of models are the corresponding linear combination of Shapley values of the underlying models.

Decomposing model disparity into feature contributions ignores the possible causal relationships between features. To tackle this problem, FACTS (Fairness-Aware Causal paTh decomposition) [84] proposes a framework that measures the influence $\Phi_{f(X)}(p_i)$ of a causal path p_i (instead of a feature i) linking the protected attribute with the outcome variable. FACTS decomposes the model disparity by replacing the feature i with the causal path p_i in the disparity measure $D\Phi_f(i)$. Fairness

is improved by using an optimization function to exclude the causal paths that contribute the most to the disparity.

Discussion: Shapley-based approaches can be computationally expensive, especially when explanations must be computed for many instances, or used inside optimization procedures. Nevertheless, their theoretical properties are useful for fairness analysis. For example, due to the additivity property, the total disparity can be decomposed into feature-level contributions, identifying which attributes drive unfairness [83].

The proposed approaches connect explanations with fairness in several ways. Some focus on the influence of the protected attribute [79], [81], while others also consider the contribution of proxy attributes [82]. They address different fairness notions: some relate attributions to distributive fairness metrics, such as demographic parity [81], [83], whereas others use attribution similarity to assess procedural fairness [80]. Feature-based methods attribute unfairness to variables, while causal-path methods such as FACTS decompose disparity along paths from the protected attribute to the prediction, providing a more structural view but relying on the assumed causal graph [84].

Another distinction concerns the level of comparison. Some methods aggregate local attributions across demographic groups [79], [81], [83], although this may hide individual variation. To address this, some compare full attribution distributions rather than means [79], [80], while others compare explanations of similar individuals from different groups [80].

C. Explanations Using a Data-Based Approach

A different approach to explaining unfairness is through data-based explanations [48], [85], [86], which aim to identify the instances in the training data that, if removed or modified, would lead to the greatest reduction in classification bias.

Gopher [48], [85], introduces the concept of causal responsibility to quantify the impact of these data interventions and uncover the root causes of model bias. Rather than focusing on individual instances, Gopher identifies compact, interpretable patterns, conjunctions of feature predicates, that describe coherent subpopulations in the training data most responsible for unfair outcomes. To ensure meaningful and non-redundant explanations, Gopher selects patterns with high responsibility scores while enforcing diversity among them. It approximates the impact of data modifications using second-order influence functions and gradient-based techniques, enabling efficient estimation without costly model retraining. Furthermore, Gopher employs a lattice-based search strategy, enhanced by pruning heuristics based on pattern support and influence, to effectively navigate the large space of potential explanatory patterns.

Similarly, FUME [86] proposes a model-agnostic framework to explain fairness violations by estimating how model fairness changes when specific parts of the training data are removed. Fairness violations are defined in terms of standard metrics such as statistical parity and equalized odds, and a training subset is considered attributable to a violation, if its removal improves fairness on the test set. Instead of retraining the model for every possible subset FUME leverages insights from the field of machine unlearning, where the goal is to “forget” specific

training instances without retraining from scratch. To manage the exponential number of potential training subsets, FUME organizes the subsets in a hierarchical lattice structure inspired by the apriori algorithm used in frequent itemset mining. This structure allows the method to efficiently search for the top- k predicate-based subsets most responsible for the violation. Several pruning strategies are applied during the search to eliminate irrelevant or low-impact subsets.

Discussion: Data-based approaches explain unfairness by identifying influential parts of the training data rather than attributing bias to features. This makes them actionable, since they point to subsets of the data that can be inspected or modified. However, they depend on the chosen fairness metric and intervention-effect estimates. Computing the exact impact of removing or modifying every candidate subset would require repeated model retraining, which is computationally expensive. Therefore, these approaches rely on approximations, such as influence functions, machine unlearning, and pruning strategies, which may affect the quality of the explanations.

D. Explanations Using Counterfactuals

Given a prediction model f and a data point $\mathbf{x}$, a *counterfactual explanation* is a similar data point $\mathbf{x}'$ for which the prediction changes, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. This is typically formulated as the following optimization problem [45]:

$$\begin{aligned} \mathbf{x}' &= \arg \min \text{distance}(\mathbf{x}, \mathbf{x}') \quad \text{s.t.} \quad f(\mathbf{x}') \neq f(\mathbf{x}) \\ \mathbf{x}' &= \mathbf{x} + \delta, \end{aligned}$$

where δ is the perturbation applied independently to the factual instance $\mathbf{x}$ to obtain its counterfactual counterpart $\mathbf{x}'$. In addition to minimizing distance, other commonly used desiderata for generating counterfactuals include sparsity, i.e., changing only a small number of features; actionability, e.g., avoiding changes to immutable features such as race; and plausibility, i.e., ensuring that suggested changes respect correlations among features [29].

The basic idea of employing counterfactuals to explain fairness for an individual $\mathbf{x}$ mapped to the negative class ($f(\mathbf{x}) = 0$) is to generate a counterfactual $\mathbf{x}'$ such that $\mathbf{x}'$ is mapped to the favorable class ($f(\mathbf{x}') = 1$). At the group level, this is extended by generating counterfactuals for the negatively classified instances in each group.

Unlike other explainability techniques, such as LIME and Shapley values, which explain why the model made a specific prediction, counterfactual explanations additionally suggest feature changes required to achieve a desired outcome. For example, consider an individual $\mathbf{x}$ whose loan request has been denied. The feature changes indicated in the corresponding counterfactual explanation $\mathbf{x}'$ can provide feedback to help the individual modify their features and transition to the favorable side of the decision boundary, i.e., to obtain the loan.

In the context of fairness, this property of counterfactuals has been used to provide a more elaborate definition of unfairness that incorporates the cost of mitigating it. For a specific instance $\mathbf{x}$, this cost, often referred to as *burden*, is quantified by the distance of $\mathbf{x}$ from its counterfactual $\mathbf{x}'$ [87]. The burden for a group G is measured by the expected distance between the

instances in the group and their corresponding counterfactuals, reflecting the extent of change needed to alter the decision of the model for the group:

$$\text{Burden}(G) = \mathbb{E}_{\mathbf{x} \in G}[\text{distance}(\mathbf{x}, \mathbf{x}')],$$

where $\mathbb{E}$ denotes expectation. The larger the burden of a group, the more unfair the model towards the group. Burden can be viewed as a weighted version of demographic parity, as it captures not only the proportion of individuals assigned to the negative class, but also the difficulty of reaching a favorable outcome, measured by the distance to the decision boundary [87]. Recent work uses burden to quantify the cost incurred in achieving fair clustering, capturing both individual-level and group-level costs [88].

Besides demographic parity, other types of group fairness can be captured by generating counterfactuals for appropriate subsets of each group. In particular, focusing on error-based fairness, the authors of [89] generate counterfactuals for the subset $FN(G)$ of the instances of the group G falsely mapped to the negative class. In this case, the burden captures the cost of reversing the decision for the unfavorably misclassified instances. Let $FNR(G) = P(\hat{y} = 0 \mid y = 1, \mathbf{x} \in G)$ be the false negative rate and $FN(G) = \{\mathbf{x} \in G \mid y = 1, \hat{y} = 0\}$ be the set of false negatives in group G . The *accuracy weighted burden* (AWB) for a group G is defined as:

$$\text{AWB}(G) = FNR(G) \mathbb{E}_{\mathbf{x} \in FN(G)}[\text{distance}(\mathbf{x}, \mathbf{x}')].$$

Counterfactuals have also been used to detect biases and understand the causes of unfairness. Assume two groups: the protected group G^+ and the non-protected group G^- . PreCoF (Predictive Counterfactual Fairness) [90] generates counterfactuals for the instances of G^+ and G^- that are mapped to the negative class. It then compares the relative frequency of the attributes whose values have been changed in the counterfactuals generated for each of the groups. The objective is to determine how much more frequently a certain attribute contributes to a negative outcome decision (i.e., as part of the counterfactual explanation) for members of the protected group compared to members of the non-protected group.

Specifically, for a feature i and instance x mapped to the negative class, PreCoF tests whether changing the value of i to one of a set of default values would result in mapping x to the favorable class. Then it computes the percentage of instances changing class for individuals in the protected class, denoted $CoF(i, s)$ and for individuals in the non-protected class, denoted $CoF(i, ns)$, and computes the difference:

$$\text{PreCoF}(i) = CoF(i, s) - CoF(i, ns).$$

The features i with the largest $\text{PreCoF}(i)$ values are considered the ones most responsible for the unfairness of the model. A distinction is also made between detecting *explicit bias*, where the model directly uses the protected attribute, and *implicit bias*, where the model does not directly use the protected attribute but relies on other proxy attributes. Explicit bias is tested by searching for counterfactual explanations that consist solely of changes of the protected attribute.

Note that counterfactual explanations are not directly related to the type of fairness called counterfactual fairness [12]. A key distinction is that counterfactual explanations focus on revealing the changes needed for different classification outcomes, while counterfactual fairness aims to assess fairness by considering predictions under different demographic scenarios using causal models [90].

1) *Actionable Recourse*: An issue with counterfactuals is that it is unclear whether the suggested actions can be used to design mitigation actions. Specifically, the suggested shifts in feature values may not be feasible. Additional constraints could be considered, for example, avoiding updates of immutable attributes [91]. Furthermore, there may be causal dependencies between attributes that need to be taken into account.

To address this problem, instead of interpreting actions as independent feature manipulations, *actionable recourse* [50] suggests that actions should be interpreted as interventions, termed *flipsets*, in a causal model. Given a structural causal model (SCM), actions are modelled as structural interventions, which can be thought of as transformations between SCMs. The following optimization objective is formulated:

$$\begin{aligned} A^* \in \arg \min_{A \in \mathcal{A}} \text{cost}(A; \mathbf{x}) \quad & \text{subject to } f(\mathbf{x}') \neq f(\mathbf{x}) \\ \mathbf{x}' = F_A(F^{-1}(\mathbf{x})) \\ \mathbf{x}' \in \mathcal{P}, \quad A \in \mathcal{F}, \end{aligned}$$

where A is the set of interventions that can be constructed as $A = \text{do}(\{X_i := a_i\}_{i \in I})$, the “do” operator replaces the structural equation in the SCM when it is applied to a variable X_i , $A^* \in \mathcal{F}$ is the minimal set of actions for the minimally costly recourse, $\text{cost}(\cdot; \mathbf{x}_F) : \mathcal{F} \times X \rightarrow \mathbb{R}^+$ is the cost function and $\mathbf{x}' = F_{A^*}(F^{-1}(\mathbf{x}))$ is the resulting structural counterfactual and $\mathcal{P}$ refers to plausibility constraints.

In [47], a distinction is made between *contrastive explanations* that are explanations used to understand unfairness (e.g., what input changes would lead to a favorable outcome) and *consequential recommendations* used to mitigate unfairness (e.g., actions would also lead to the specific changes). A counterfactual-based explanation is the nearest contrastive explanation and an actionable recourse, a consequential recommendation.

Another approach to incorporating causality is offered by the *probabilistic contrastive counterfactuals* that exploit probabilistic causal models [92]. Actions are in the form of intervention queries. Unlike actionable recourse, intervention queries can be estimated from historical data, thereby making no assumptions about the structural equations in the underlying probabilistic causal models.

2) *Group Counterfactuals*: Counterfactual explanations defined for groups of instances are referred to as group counterfactuals. Defining and generating group counterfactuals introduces novel challenges. The problem was first introduced in [93], where it was modeled as a constraint optimization problem. At a high level, group counterfactuals consist of conditions on feature values, called *actions*, that the instances in the group must satisfy to receive a positive outcome. Recent work includes (a)

GLOBE-CE that defines a group counterfactual to be a global direction along which a group of instances may travel to alter their predictions [94], (b) counterfactual explanation trees that build decision trees by assigning actions to multiple instances simultaneously [95], and (c) FACTS that uses a frequent itemset approach [96].

FACTS [96] (Fairness-Aware Counterfactuals for Subgroups) introduces a framework for auditing fairness using group-level counterfactuals. FACTS systematically explores the space of possible actions A and evaluates their impact at the group level. For each action $a \in A$ and group G , the method computes the recourse cost $rc(A, x)$, which reflects the effort required by an individual instance x to achieve a favorable outcome. It also assesses the effectiveness of each action, defined as $\text{eff}(a, G)$, i.e., how many individuals in group G successfully achieve recourse through a . Based on this, FACTS detects potential recourse bias between protected and non-protected subgroups. To evaluate fairness, FACTS introduces criteria such as *equal effectiveness*, which requires that the proportion of individuals achieving recourse be the same in both the protected group G^+ and the non-protected group G^- , and *equal choice of recourse*, which requires that both groups have access to the same number of sufficiently effective actions, i.e., those that help at least $100\phi\%$ of the group, for some $\phi \in [0, 1]$.

FACEGroup [97] (Feasible and Actionable Counterfactual Explanations for Group Fairness) generates actionable group counterfactuals using a density-weighted feasibility graph, where nodes represent data points and edges denote feasible transitions. The feasibility graph naturally partitions data into weakly connected components, forming (sub)groups for generating counterfactuals. Leveraging this structure, FACEGroup develops algorithms to produce a compact yet representative counterfactual set and introduces novel fairness metrics. These metrics enhance fairness evaluation by capturing trade-offs in counterfactual generation at both group and subgroup levels. FACEGroup proposes two types of fairness metrics: attribution-based ones that measure how frequently features change across groups, and burden-based metrics. The key factors in these metrics are the number of counterfactuals k for a group X , the maximum cost d an individual of a group must incur to change their outcome using the proposed set of counterfactuals S , and coverage c which is the percentage of individuals who successfully achieve the desired outcome using the proposed counterfactuals. Some key metrics include:

- *Minimum k (k_0) and cost (d_0) for full coverage.*

$$k_0 = \min\{k \mid \exists S, |S| \leq k, \text{coverage}(X, S) = |X|\},$$

$$d_0 = \min\{d \mid \exists S, \text{cost}(X, S) \leq d, \text{coverage}(X, S) = |X|\}.$$

- *AUC-based fairness measures*: These quantify the trade-off between cost, coverage, and number of counterfactuals across a range of values, avoiding bias from fixed parameters. One example is kAUC, which measures how efficiently a group can achieve coverage across a range of cost values for a given number of counterfactuals.

$$kAUC(k) = \int_{d_{\min}}^{d_{\max}} \text{coverage}(X, S_{k,d}) dd,$$

where:

$$S_{k,d} = \operatorname{argmax}_{|S| \leq k, \operatorname{cost}(X,S) \leq d} |\operatorname{coverage}(X, S)|.$$

This metric is complemented by the saturation point $sp(k)$, defined as the minimum cost at which the maximum possible coverage is achieved for a given k .

Discussion: Overall, counterfactual-based approaches provide a complementary perspective to fairness explanations by relating unfairness to the difficulty of crossing the decision boundary. They can be used to quantify recourse burden [87], [89], identify features that act as barriers for protected groups [90], and compare the cost and effectiveness of actions across groups [96], [97]. However, these conclusions depend strongly on the choice of cost function used to measure the effort required for individuals to achieve favorable outcomes, and group-level summaries may hide individual differences.

A key distinction is between counterfactual explanations and actionable recourse. While standard counterfactuals identify nearby instances with different predictions, recourse methods require feasible changes [50], [97]. However, this feasibility depends on predefined actionable features or causal constraints.

E. Explaining Fairness for Graphs

Next, we discuss domain-specific challenges, focusing on graph models for classification, recommendation and ranking tasks. Graph Neural Networks (GNNs), in particular, have become dominant for graph classification and recommendation due to their strong performance, but they risk yielding biased results against certain demographics [98]. Explaining such biases is critical for safe deployment. Even non-GNN models on graphs (like knowledge-graph-based recommenders) exhibit fairness issues. For example, explainable recommenders can unfairly favor “active” users with more data, while “inactive” users get poorer recommendations [99]. Most work in graph fairness explainability has focused on the example-based [98], [100], [101], [102], [103], [104], rather than feature-based [99] approaches.

Feature-based explanations aim to pinpoint which node attributes or structural characteristics (e.g., degree, community membership, homophily levels) correlate with model bias. In [99], the authors analyzed path-feature distributions and discovered that users with fewer interaction paths in a knowledge-graph recommender system received disproportionately poorer recommendations. Because inactive users had fewer paths available, the model relied heavily on data from more active users, introducing cross-user bias. To address this, they applied a post-hoc fairness-aware re-ranking heuristic, tuning utility-fairness trade-offs while preserving path-based explanations without retraining the KG reasoning recommender.

Example-based explanations diagnose unfair decisions by reference to specific instances or perturbations of the data, for example, pointing to particular nodes/edges or hypothetical modifications that alter the fairness outcome. This category includes counterfactual explanations (showing how changing part of the input graph would affect fairness) and influence-based

explanations (identifying which existing data points most contributed to the bias). The work in [98] proposed a post-hoc structural explanation framework that, for any target node, jointly optimizes two sets of input graph edges as a paired bias/fairness explanation: one set that maximally accounts for the biased outcome of the model, and another that maximally contributes to a fair outcome for that node. The method identifies which connections in the graph are causing an individual prediction to be unfair, and which connections, if present, would make it fair. These edge sets serve as a counterfactual explanation: by removing or adding those relationships, the output of the model would become more fair or unfair.

The work in [100] shifts the focus to the training data. Instead of explaining a single prediction by its input features, this post-hoc method attributes model unfairness to specific training nodes. The authors define a new fairness metric called Probabilistic Distribution Disparity (PDD) to quantify the bias in a GNN’s predictions across demographic groups, and then they develop a closed-form influence approximation formulation to estimate each training node’s influence on this bias without per-node retraining. In effect, the explanation for the overall unfair behavior of the model is given as a ranking of influential training examples (nodes) that are most responsible for causing the biased outcomes. This is an example-based, data-centric explanation: it tells us which specific nodes in the training graph led the GNN to become biased. The authors of [101] propose a counterfactual-based explainer that finds minimal counterfactual perturbations to the user–item interaction graph. The recommendation scenario is represented as a bipartite graph (users and items as nodes, interactions as edges). Given a measured utility disparity between a protected group of users and others, their algorithm searches for a small set of user–item edges to remove such that the disparity (unfairness) is significantly reduced. Intuitively, the identified edges are those connections in the training data that were causing the recommender’s bias against the protected group. By perturbing the graph structure (removing or possibly rewiring certain interactions), the method produces a counterfactual explanation, while structural-pattern analysis of the perturbed edges surfaces regularities such as popularity and homophily behind the unfairness.

A complementary strategy is to use counterfactuals not just to explain bias, but to actively mitigate it. The work in [102] introduces a graph augmentation approach guided by counterfactual explanations that inverts the usual perturbation direction. The idea is to identify new user–item interactions (edges in the bipartite graph) that would make the recommendations fairer if those edges existed, and then add these synthetic connections into the data creating “what-if” interactions that explain the original unfairness by their absence and simultaneously serve to reduce that unfairness when added. Several very recent works use example-based explanations to analyze fairness at a finer structural level in GNNs. The authors of [103] focus on the impact of individual graph edges. They propose a framework to quantify how much each edge contributes to fairness or unfairness in a GNN’s predictions. In practice, this involves repeatedly measuring changes in a fairness metric (e.g., group fairness disparity) when each edge is removed in turn (a counterfactual

removal test). The output is a per-edge fairness-importance score with respect to the fairness of the model – essentially a feature-attribution explanation at the level of graph structure. Finally, the work in [104] formulates a multi-objective evolutionary optimization problem that keeps compactness, diagnostic faithfulness, and actionability as separate objectives rather than collapsing them into a single regularized loss: they search for a small subgraph (a set of nodes and edges) that is highly responsible for the discriminatory decision of a model, while also being as concise as possible and indicating a route to fix the issue.

Discussion: Graph-based fairness explanations attribute bias to structural elements: edges [98], [101], [102], [103], training nodes [100], subgraphs [104], or KG reasoning paths [99], which makes them actionable: the identified elements can be removed, added, or re-weighted to mitigate the disparity. Approaches differ in the targeted fairness notion (group-fairness disparities [98], [103], [104], or consumer-side utility disparity across demographic [101], [102] or activity-defined [99] groups) and on whether they are primarily diagnostic [98], [100], [103] or coupled with a remediation step [99], [102], [104]. A common limitation is that attributions inherit the non-IID propagation of graphs and are thus sensitive to the chosen neighborhood and message-passing scheme.

F. Explaining Fairness for Recommendation and Ranking Tasks

Recommendation explanations can be feature-based, example-based, or approximation-based. In a fairness context, however, feature-based approaches face additional challenges [107], shifting the focus to counterfactual example-based methods that can handle sparse data more effectively [107], [108], [109], [110]. Traditional feature-based fairness explainability has instead been applied mainly to ranking tasks [111].

Feature-based approaches for explaining fairness in ranking [112], [113] identify bias based on the concept of representation fairness in rankings, which requires that subgroups defined by specific attribute-value combinations appear proportionally in the top-ranked positions, reflecting their overall representation in the dataset. The work in [112] identifies which attributes significantly affected the ranking of the biased group using Shapley values, under both global representation bounds and proportional representation. Extending this line of work, DEXER [113] packages this Shapley-based analysis into an interactive end-user system for exploratory bias auditing, showing the user visualizations of those features' value distributions for the biased group versus the top- k items.

Approximation-based explanations for recommendation tasks have been studied in [111], which uses an explanatory surrogate model to examine various data characteristics (e.g. user count, item popularity distribution, rating sparsity) and how these correlate with fairness and accuracy metrics across collaborative-filtering algorithms. In their approach, the authors define several dataset-level features as explanatory variables and measure fairness outcomes (MAD-based consumer disparity by

gender or age) as dependent variables. They then fit regression models and perform statistical significance tests to identify which factors significantly impact fairness. For instance, they might find that "item popularity bias (long-tail length)" has a significant negative effect on a fairness metric, whereas "user activity level variance" does not, thus explaining that popularity imbalance is the primary driver of unfairness in that recommender.

Example-based approaches comprise mainly counterfactual-based works such as the work in [107] that presents Counterfactual Explainable Fairness (CEF), a framework for explaining unfairness in recommender systems by finding the minimal changes to input features that would make the recommendation results fairer (more balanced exposure) without significantly hurting accuracy. For example, CEF might determine that slightly increasing the relevance of a genre feature for certain users would raise the exposure of less-popular items, improving fairness. Technically, the authors formulate an optimization problem: they learn feature perturbations that achieve a specified fairness improvement under a constraint on utility loss. By solving this problem, CEF assigns each feature a counterfactual-derived explainability score and produces a ranked list of critical features along with how changing them affects the fairness-utility trade-off. The top features form the fairness explanation (e.g. "user age and region are the main drivers of exposure disparity"). In addition, the authors of [108] propose CFairER, a Counterfactual Fairness Explanation model for recommendations that leverages a knowledge-aware approach. This method generates discrete attribute-level counterfactual explanations for item ranking fairness, particularly focusing on item exposure unfairness (some items receiving systematically less exposure). CFairER represents users and items in a heterogeneous information network (HIN) that includes real-world attributes (e.g. item categories, user demographics) and uses an off-policy reinforcement learning algorithm with attentive action pruning to search for the minimal set of attribute changes that would improve the model's fairness metric, avoiding CEF's gradient/continuous-score formulation. For example, the system might find that if a certain niche item attribute (say, "author nationality") were given slightly more weight or a user's profile were augmented with an interest tag, the recommender would give more exposure to items from an under-represented group, thus reducing disparity.

The work in [109] approaches the problem from a different angle, the perspective of individual users or items. They introduce Adding-Based Counterfactual Fairness Reasoning (ACFR) to explain how specific users or items contribute to overall unfairness. While traditional "erasing"-based analysis (removing a user or item) can break the collaborative filtering structure, ACFR instead adds hypothetical new interactions for a user or item to see how it would affect fairness, adapting counterfactual reasoning to pure collaborative-filtering settings in which no item/user features are available. For example, to explain a popularity bias, ACFR might simulate a cold-start user suddenly interacting with a long-tail item and observe if the fairness metrics improve – if so, it means that user's lack of interactions was partly causing unfairness. The explanations take

the form of which additional interactions would have needed to exist to make the system fairer, highlighting the roles of particular users or items in the fairness outcome. Concluding the counterfactual line of works, the work in [110] extends counterfactual reasoning from individual instances to groups, applying counterfactual edge removal in a random-walk-based recommender (RecWalk [114]) to investigate the minimal rating changes needed to correct biases in user or item groups and provides an edge-centric, concise group-level perspective for bipartite user–item graphs, quantifying how each removed edge influences group-level fairness metrics in the final recommended lists, with explanations at both the user and item levels.

Discussion: Feature-based methods are mostly confined to ranking, where attribute-value patterns and Shapley values reveal which features drive biased group representation [112], [113]. In recommendation, sparse categorical inputs shift the focus to counterfactual methods perturbing features [107], [108] or interactions [109], [110], while approximation-based surrogates explain how dataset characteristics drive fairness–accuracy trade-offs across collaborative-filtering algorithms [111]. Methods further differ in the audited side (item-side exposure [107], [108], user- and item-side utility [109], [110], or proportional representation [112], [113]) and in granularity (individual, group, or dataset level). A common limitation is that outputs are tied to the chosen recommender architecture, with no guaranteed transfer across models.

G. Caveats

The discourse on fairness in AI raises fundamental questions regarding the reliance on sensitive attributes and the validity of fairness assessments through surrogate models. In this section, we discuss five key concerns: (i) whether dependency on sensitive attributes necessarily indicates unfairness, (ii) whether explanation-based fairness claims depend on both the chosen fairness definition and the explanation method used, (iii) whether the fairness of a surrogate model faithfully reflects that of the original model, (iv) whether post-hoc explanation methods can be manipulated to mask or distort underlying model biases, and (v) whether feature-based explanations reliably improve distributive fairness in human-in-the-loop AI-assisted decision-making. These caveats underscore the importance of critically evaluating fairness claims and ensuring that explainability methods do not inadvertently serve as tools for masking unfairness rather than revealing it.

A common assumption in fairness-aware machine learning is that an algorithm should not explicitly depend on sensitive attributes such as gender or race. This perspective aligns with the notion of “fairness through unawareness”. Previous research has criticized this approach as being overly simplistic and potentially misleading, since simply ignoring sensitive attributes does not ensure fairness, as other correlated features may still encode the same information, leading to hidden biases. [116] show that disparate learning processes using the sensitive attribute only at training time can still implicitly induce treatment disparity via correlated features and within-class discrimination when the attribute is only partially encoded. Furthermore, in some contexts,

considering sensitive attributes may be necessary, e.g. to account for differential subgroup validity or to counteract structural disadvantages for historically marginalized groups [1].

A further caveat concerns the multidimensional nature of fairness. Fairness can refer to different notions, such as distributive fairness, procedural fairness, individual fairness, or group fairness. Consequently, fairness assessments may lead to different conclusions depending on the specific notion being evaluated [4]. This also applies to explanation-based fairness analysis, since explanations may be used to assess different aspects of fairness, such as whether specific features contribute to violations of distributive fairness across groups, or whether the model relies on appropriate features. However, these dimensions are not interchangeable, so relationships such as procedural versus distributive fairness require further study [73], [117]. In addition, explanation-based fairness conclusions may depend on the chosen explanation method. This concern is related to the *disagreement problem*, where different post-hoc explainers, such as LIME or SHAP, may produce inconsistent feature rankings for the same prediction [118].

Another significant concern is the extent to which surrogate models often used for explainability, faithfully represent the fairness properties of the original black-box model. Since post-hoc explanations often simplify complex models, they may provide incomplete or misleading insights, especially when used to assess fairness-sensitive decisions [119]. This mismatch can lead to fairwashing, where a surrogate explanation appears fairer than the underlying black-box model and therefore gives a misleading impression of the fairness of the model [1], [120]. A follow-up [121] quantifies the fidelity–unfairness trade-off and shows that fairwashed explanations transfer across models, hindering audit-based detection. Consequently, reliance on surrogate explanations for fairness assessments requires caution, as they may fail to uncover underlying biases.

A related line of research underscores how post-hoc explanation methods can be manipulated to mask or distort underlying model biases, potentially introducing fairness disparities. [122] demonstrate that widely used black-box explanation tools such as LIME and SHAP can be “fooled” to conceal discriminatory model behavior in tabular classification tasks. Recent work further shows that this vulnerability is not limited to explicit adversarial attacks, [123] show that SHAP-based explanations are highly sensitive to feature representation, such that common preprocessing operations, including bucketizing age or regrouping race categories, can substantially reduce the apparent importance of protected attributes without necessarily changing the underlying prediction behavior. Consequently, explanation-based fairness assessments may be vulnerable not only to deliberate manipulation, but also to seemingly benign data engineering choices. Hence, designing explanation methods that remain faithful under adversarial and representation-sensitive conditions is crucial for ensuring that explanation-based fairness assessments are trustworthy.

Finally, another caveat concerns whether feature-based explanations reliably improve distributive fairness in human-in-the-loop AI-assisted decision-making. Although explanations are often expected to help humans identify and correct unfair AI

TABLE III

OVERVIEW OF APPROACHES FOR FAIRNESS OF EXPLANATIONS. **MODALITY (MOD.):** TAB = TABULAR, GR = GRAPH. **ACCESS (ACC.):** ● BLACK-BOX, ○ GRADIENT ACCESS, ○ WHITE-BOX. **AGNOSTIC (AGN.):** ✓ MODEL AGNOSTIC, ✗ MODEL SPECIFIC. **COVERAGE (COV.):** ● GLOBAL, ○ LOCAL. **LEVEL:** ● INDIVIDUAL, ○ GROUP. **CRITERIA:** OBS. = OBSERVATIONAL, CAUS. = CAUSAL.

Appr.	Mod.	Explanation				Fairness			
		Stage	Acc.	Agn.	Cov.	Type	Level	Type	Criteria
[127]	Tab	Post	●	✓	○	CFEs	●	Distance-based	Obs.
[105]	Tab	Post	●	✓	●	Recourse	○	Fairness of recourse	Obs.
[106]	Tab	Post	●	✓	○	Recourse	○	Fairness of recourse	Caus.
[126]	Tab	Post	●	✓	○	Feature-based + Surrogate-based	○	Procedural	Obs.
[125]	Tab	Post	●	✓	○	Surrogate-based: Subgroup fidelity gap	○	Procedural	Obs.
[130]	Gr	Post	●	✓	○	CFEs + Feature-based	○	Counterfactual	Obs.
[131]	Tab	In	○	✗	○	Feature-based: Emotion-balanced textual explanations	○	Base-rates	Obs.
[99]	Gr	Post	●	✓	○	Example-based: Top-k KG paths	○	Constraints + Consumer	Obs.
[132]	Gr	Post	●	✓	○	Example-based: Summary subgraph	○	Constraints + Producer	Obs.

recommendations, [124] show that they may instead shape fairness perceptions and reliance behavior of users without necessarily helping them distinguish correct from incorrect recommendations. Users may be more likely to follow recommendations from an AI system they perceive as fair, and override those perceived as unfair, regardless of whether the recommendation is actually correct.

IV. FAIRNESS OF EXPLANATIONS

On a different front, explanation methods are not free from biases and, therefore, must follow fairness criteria. Consequently, pursuing fair explanations in machine learning and decision-making systems has become a pivotal area. Fair explanations aim to ensure that explanations are free from biases, capture model disparities correctly, and maintain parity across different demographic groups. Fair explanations try to limit the effect of data and algorithmic unfairness exclusively at the level of explanations, which manifests by different quality or diversity in the explanations generated between groups or individual users. Table III summarizes these approaches along our explanation and fairness dimensions.

Several approaches explore the quality and fairness of explanations across different demographic groups. In [125], the authors investigate the fairness of explanations using surrogate models by introducing the notion of fidelity. Fidelity is defined as how well an explanation model $\mathcal{E}$ approximates the behavior of a black-box model B :

$$\text{Explanation Fidelity} = \frac{1}{N} \sum_{i=1}^N L(B(x_i), \mathcal{E}(x_i)),$$

where x_i is an instance from the dataset, N the total instances and L is a performance metric like accuracy, mean error, etc. They show that the gap in demographic parity between $\mathcal{E}$ and B is bounded by the difference in the mean residual error of data points in each group. Based on the fidelity definition, they form two metrics, *maximum fidelity gap from average*, which measures the extent to which the fidelity of the disadvantaged group is lower than the average, and *mean fidelity gap among subgroups*, which measures how much the fidelity differs across

subgroups. They observe that the fidelity gaps are largest for the least-fair black-boxes. However, even when training fair models, fidelity gaps are still observed. To reduce fidelity gaps across groups, they propose not only robust training but also data-distribution-aware training methods that leverage causal knowledge.

Similarly, [126] proposes several metrics to assess fairness in explanations. They introduce various measures, including *prediction gap fidelity*, which evaluates how much the prediction of a model changes when alterations are made to less important features in the explanation, *instability*, that measures the dissimilarity of explanations for similar data points, *inconsistency*, that measures the dissimilarity of repeated explanations for the same point, and *complexity*, assessing the ease of understanding an explanation by users. These are defined as follows, where f denotes the model being explained, $\mathcal{E}$ the explanation method, $\mathbb{E}[\cdot]$ the expectation operator, and $h(\mathbf{x})$ the predicted class probabilities of f at $\mathbf{x}$:

$$\text{Pred.gap}(\mathbf{x}, f, \mathcal{E}) = \mathbb{E}[|h(\mathbf{x}) - h(\tilde{\mathbf{x}})|],$$

where $\tilde{\mathbf{x}}$ is a modified version of $\mathbf{x}$ with noise added to less important features.

$$\text{Instability}(\mathbf{x}, f, \mathcal{E}) = \mathbb{E}[|\mathcal{E}(\mathbf{x}, f) - \mathcal{E}(\tilde{\mathbf{x}}, f)|_1],$$

where $\tilde{\mathbf{x}}$ is a slightly noisy version of the input data point $\mathbf{x}$.

$$\text{Inconsistency}(\mathbf{x}, f, \mathcal{E}) = \frac{1}{m} \sum_{j=1}^m \|\mathcal{E}_0(\mathbf{x}, f) - \mathcal{E}_j(\mathbf{x}, f)\|_1,$$

where m is the number of repeated runs, $\mathcal{E}_0$ is a reference explanation, and $\mathcal{E}_j$ is the explanation generated with random seed j , and

$$\text{Complexity}(\mathbf{x}, f, \mathcal{E}) = \sum_i^d \mathbb{1}[w_i > t],$$

where $\mathbf{w} = \mathcal{E}(\mathbf{x}, f)$ is the local explanation vector, w_i represents the importance score of the i -th feature, d is the number of features, t is a threshold value, and $\mathbb{1}[w_i > t]$ is an indicator function that equals 1 if the importance score of the i -th feature is greater than t , and 0 otherwise. They assess the disparity of

the quality of explanations per metric if they result in significant differences between groups.

Also, [127] examines the fairness of explanations by investigating the robustness of counterfactual explanations. Counterfactual explanation methods can exhibit instability with respect to small changes in the input, resulting in significantly different explanations for similar individuals. This instability can lead to the unfair treatment of individuals who are otherwise similar. To address this, x_{cf} is defined as the counterfactual explanation for the original instance x_{orig} . They also consider $x \sim p(x_{orig})$ as a perturbed version of x_{orig} drawn from a probability density p , and they analyze its corresponding counterfactual explanation. The local instability of the counterfactual explanation is measured as the expected distance between the counterfactual explanation for the original instance and that for the perturbed sample. This distance quantifies how much the counterfactual explanations vary with small changes in the input, thereby providing a measure of the robustness of explanations.

Finally, CounterFair introduces a counterfactual explanation algorithm that aims to equalize burden across groups, including subgroups discovered through shared counterfactuals [128], [129].

A. Fair Recourse

Recourse has been employed to uncover discriminatory patterns within machine learning models against individuals and groups. This is accomplished through the implementation of recourse actions. For instance, if a woman is compelled to undergo increasingly demanding adjustments and efforts to secure a job prediction, in contrast to the actions recommended for a man, it signifies a bias in favor of men within the model. In certain scenarios, demographic parity - equality in positive predictions for each sensitive group - may be maintained, signifying equality in positive predictions concerning the sensitive attribute. However, if the recourse actions required to move from a negative to a positive outcome differ more substantially for one group than another, we observe *unfair recourse*. The objective then shifts from equalizing positive predictions to equalizing recourse actions across groups.

In this manner, [105] focuses on optimizing recourse across demographic groups. They employ a regularized objective while training the classifier to ensure an equal average distance to the decision boundary for different groups. This approach is adaptable to both linear and non-linear classifiers. Another approach is proposed by [106], where the authors argue that recourse-based fairness considerations should be grounded in a causal model, capturing the effects of real-world interventions where features are often causally related. They propose using a Structural Causal Model (SCM) to model causal relationships between protected attributes and other features. They introduce two concepts: individually fair causal recourse, where the cost of recourse would be the same if the individual belonged to a different protected group, and group-level fair causal recourse, which ensures equal recourse across groups. In both concepts, the cost of recourse is calculated within a causal model. The authors then use constrained optimization to train a classifier

that maintains causal recourse fairness at either the group or individual level.

B. Fair Explanations for Graphs

Ensuring fairness in graph explanations, particularly GNNs, involves aligning the graph model interpretability with fairness principles. However, developing fair explanations for graphs poses unique challenges due to the relational and structural complexities inherent in graph data. Recent advancements have begun to lay the groundwork for fair explanations in graph-based models. One notable contribution is [130], which introduces GraphXAI, a framework for benchmarking GNN explanation methods that incorporates fairness as a core metric in its evaluation protocols. Using bias-tunable datasets, the framework assesses group (consistent quality across groups) and counterfactual fairness (unchanged when sensitive attributes are flipped).

C. Fair Explanations for Recommendation

Recent work examines whether the content of recommender explanations is fair across user and item sub-groups, complementing traditional studies of fairness in the recommendations themselves. For textual explanations, the authors of [131] show that state-of-the-art generators overuse the “happy” emotion while under-representing others, such as “neutral” or “sad,” leading to an unbalanced emotion distribution. They introduce an emotion-aware regularization term (alongside a fairness loss) to rebalance these proportions and warn that an exaggeratedly positive tilt could erode user trust and perceived transparency. For knowledge-graph path explanations, the authors of [99] measure how diverse the explanation paths are for different user groups. Using Simpson’s Index of Diversity (SID), they find that inactive users (with few interactions) receive markedly lower SID scores than active users, which is evidence that some sub-groups are deprived of explanation variety. Their FairKGRec framework introduces a fairness-aware objective that raises SID for the disadvantaged group while maintaining recommendation quality. Similarly, the authors of [132] propose path-based summary explanations that produce a unified rationale per user or item, as well as per user/item groups, in a knowledge graph recommender. Treating explanation-length consistency as a fairness proxy, they show that standard per-item paths are much longer (and thus harder to grasp) for long-tail items, whereas their summaries remain short and uniform across popularity groups.

V. SUMMARY AND FUTURE DIRECTIONS

In this paper, we first developed taxonomies for fairness and explanation methods, categorizing existing literature based on key dimensions. We then explored the intersection of fairness and explainability. Table III provides a comprehensive comparison of the various approaches designed to ensure fairness in explanations. Tables I and II present a comprehensive comparison of the various approaches developed to explain fairness in classification and recommendation/ranking tasks, respectively, and can serve as a detailed guide for understanding the landscape

of fairness explanation methods, highlighting their diversity in methodology and objectives.

In terms of *explanations* (see Fig. 2), Tables I and II categorize the data modality considered by each approach (e.g., tabular or graph), whether approaches utilize inherently explainable models, explain the data before model training, or explain non-transparent models (Post). It further specifies their access type, whether model-agnostic or model-specific, determining applicability across models. In terms of coverage, it indicates whether explanations offer a holistic view or case-specific insights. Then, the explanation type (e.g., CFE, Shapley values) and output format (e.g., structural change, edge set) of each approach are listed, providing insights into how explanations are presented.

In terms of *fairness* (see Fig. 1), Tables I and II differentiate between individual and group-level fairness based on sensitive attributes, and additionally indicate whether each approach follows observational or causal fairness criteria. They identify the fairness criteria each approach aims to satisfy, including statistical group fairness measures and other notions such as procedural fairness and burden. Table II further specifies the *task* for each approach, such as recommendation systems (Rec) or ranking (Rank), both tables summarize the ultimate *goal* of each method (e.g., enhancing fairness metrics, providing causal understanding, or mitigating unfairness). Two taxonomy aspects are omitted as columns for readability: Multiplicity, which is largely uniform across methods (most give one explanation per instance), and Stakeholders, which offers limited discriminative value.

A comprehensive overview of Tables I and II reveals that a predominant trend among approaches for explaining fairness is the adoption of post-processing methods, indicating a common emphasis on explaining models after training. Furthermore, most of these methods are black-box and model-agnostic. Notably, counterfactual explanations (CFEs) emerge as a prevalent technique, while the primary focus often centers around group fairness. The comparison further reveals that existing work is concentrated mainly on tabular data and observational fairness criteria, with comparatively limited attention to graph, text, or multimodal settings, as well as to causality-aware fairness explanations.

Turning to Table III, a clear trend is the predominance of post-processing techniques for fair explanations, with only a few in-processing approaches [105], [106] and no pre-processing solutions. The table further shows that most approaches focus on tabular data and observational fairness criteria, while graph-based and causality-aware fair explanation methods remain comparatively less explored. Most methods operate in a model-agnostic manner and utilize either black-box or gradient access, facilitating broad applicability across different model types. The focus remains largely on group-level fairness, with individual-level considerations being less explored. In terms of tasks, the majority of approaches are designed for classification, while a few methods address recommendation systems. These trends highlight open opportunities for developing pre-processing and in-processing solutions, advancing individual-level fairness, and

extending fair explanation techniques to tasks beyond classification.

Since most approaches target a specific type of fairness, an interesting direction is to study their applicability to other types of fairness. There is also room for exploration into dynamic fairness criteria that adapt to evolving data distributions and societal norms. This could include the development of fairness metrics and explanations that are responsive to the changing landscape of data and demographics. Furthermore, most approaches tend to focus either on individual or group levels of fairness. Therefore, future directions in this field could involve bridging the gap between individual and group, as well as subgroup levels, thus offering a more comprehensive view of fairness, akin to the advancements made by [93], [94], [97], [133]. Such mechanisms could implement hybrid models that balance the nuances of individual cases with the broader patterns observed at the group level, ensuring more equitable treatment.

Furthermore, current research is limited in its coverage of various ML tasks. While efforts have been made to explain fairness in binary classification and top-k recommendation scenarios, there is a significant gap in extending such work to more complex tasks like multiclass classification, regression, clustering, representation learning, unsupervised learning, and applications involving graphs and knowledge graphs.

Another direction is to develop methods that can generate diverse explanations, either multiple explanations of the same type or explanations of different types. Such methods could give users a broader set of interpretive and actionable resources, support better understanding, and enable more customized responses to fairness concerns. More generally, there is a need for methods and metrics that jointly capture trade-offs among utility, fairness, and explainability, so as to better assess the compromises involved in responsible algorithmic decision-making.

Finally, the majority of works focus on explanation methods aimed at understanding the causes of unfairness. There remains ample opportunity for the creation of further fairness metrics guided by these explanations and for leveraging them in the development of strategies to alleviate unfairness.

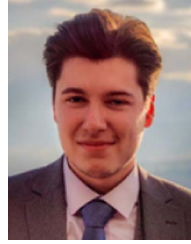
REFERENCES

- [1] L. Deck, J. Schoeffer, M. De-Arteaga, and N. Kühl, "A critical survey on fairness benefits of explainable AI," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2024, pp. 1579–1595.
- [2] C. Fragkathoulas, V. Papanikou, D. Pla Karidi, and E. Pitoura, "On explaining unfairness: An overview," in *Proc. IEEE Int. Conf. Data Eng. Workshops*, 2024, pp. 226–236.
- [3] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Comput. Surv.*, vol. 54, no. 6, 2021, Art. no. 115.
- [4] S. Caton and C. Haas, "Fairness in machine learning: A survey," *ACM Comput. Surv.*, vol. 56, no. 7, 2020, Art. no. 166.
- [5] C. Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. Durham, NC, USA: Lulu Press, 2020.
- [6] A. B. Arrieta et al., "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, 2020.
- [7] J. Zhou, F. Chen, and A. Holzinger, "Towards explainability for AI fairness," in *Proc. Int. Workshop Extending Explainable AI Beyond Deep Models Classifiers*, 2020, pp. 375–386.

- [8] E. Balkir, S. Kiritchenko, I. Nejadgholi, and K. C. Fraser, "Challenges in applying explainability methods to improve the fairness of NLP models," in *Proc. Workshop Trustworthy Natural Lang. Process.*, 2022, pp. 80–92.
- [9] R. Ramachandranpillai, R. Baeza-Yates, and F. Heintz, "FairXAI-A taxonomy and framework for fairness and explainability synergy in machine learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 6, pp. 9819–9836, Jun. 2025.
- [10] E. Pitoura, K. Stefanidis, and G. Koutrika, "Fairness in rankings and recommendations: An overview," *VLDB J.*, vol. 31, no. 3, pp. 431–458, 2022.
- [11] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proc. Innov. Theor. Comput. Sci. Conf.*, 2012, pp. 214–226.
- [12] M. J. Kusner, J. Loftus, C. Russell, and R. Silva, "Counterfactual fairness," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 4069–4079.
- [13] Y. Wu, L. Zhang, and X. Wu, "Counterfactual fairness: Unidentification, bound and algorithm," in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, pp. 1438–1444.
- [14] N. Grgic-Hlaca, M. B. Zafar, K. P. Gummadi, and A. Weller, "The case for process fairness in learning: Feature selection for fair decision making," in *Proc. NIPS Symp. Mach. Learn. Law*, 2016, Art. no. 11.
- [15] N. Grgic-Hlaca, M. B. Zafar, K. P. Gummadi, and A. Weller, "Beyond distributive fairness in algorithmic decision making: Feature selection for procedurally fair learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, Art. no. 7.
- [16] A. Castelnovo, R. Crupi, G. Greco, D. Regoli, I. G. Penco, and A. C. Cosentini, "A clarification of the nuances in the fairness metrics landscape," *Sci. Rep.*, vol. 12, no. 1, 2022, Art. no. 4209.
- [17] G. K. Patro, L. Porcaro, L. Mitchell, Q. Zhang, M. Zehlike, and N. Garg, "Fair ranking: A critical review, challenges, and future directions," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2022, pp. 1929–1942.
- [18] M. Zehlike, K. Yang, and J. Stoyanovich, "Fairness in ranking, Part I: Score-based ranking," *ACM Comput. Surv.*, vol. 55, no. 6, 2022, Art. no. 118.
- [19] M. Zehlike, K. Yang, and J. Stoyanovich, "Fairness in ranking, Part II: Learning-to-rank and recommender systems," *ACM Comput. Surv.*, vol. 55, no. 6, 2022, Art. no. 117.
- [20] Y. Wang, W. Ma, M. Zhang, Y. Liu, and S. Ma, "A survey on the fairness of recommender systems," *ACM Trans. Inf. Syst.*, vol. 41, no. 3, 2023, Art. no. 52.
- [21] D. Jin et al., "A survey on fairness-aware recommender systems," *Inf. Fusion*, vol. 100, 2023, Art. no. 101906.
- [22] Y. Deldjoo, D. Jannach, A. Bellogn , A. Difonzo, and D. Zanzonelli, "Fairness in recommender systems: Research landscape and future directions," *User Model. User-Adapted Interact.*, vol. 34, no. 1, pp. 59–108, 2024.
- [23] Y. Zhao, Y. Wang, Y. Liu, X. Cheng, C. C. Aggarwal, and T. Derr, "Fairness and diversity in recommender systems: A survey," *ACM Trans. Intell. Syst. Technol.*, vol. 16, no. 1, 2025, Art. no. 2.
- [24] Y. Dong, J. Ma, S. Wang, C. Chen, and J. Li, "Fairness in graph mining: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 10, pp. 10583–10602, Oct. 2023.
- [25] C. Laclau, C. L geron, and M. Choudhary, "A survey on fairness for machine learning on graphs," 2022, *arXiv:2205.05396*.
- [26] A. Chen et al., "Fairness-aware graph neural networks: A survey," *ACM Trans. Knowl. Discov. Data*, vol. 18, no. 6, 2024, Art. no. 138.
- [27] A. Saxena, G. Fletcher, and M. Pechenizkiy, "FairSNA: Algorithmic fairness in social network analysis," *ACM Comput. Surv.*, vol. 56, no. 8, 2024, Art. no. 213.
- [28] W. Zhang, S. Zhou, T. Walsh, and J. C. Weiss, "Fairness amidst non-IID graph data: A literature review," *AI Mag.*, vol. 46, no. 1, 2025, Art. no. e12212.
- [29] S. Verma, V. Boonsanong, M. Hoang, K. Hines, J. Dickerson, and C. Shah, "Counterfactual explanations and algorithmic recourses for machine learning: A review," *ACM Comput. Surv.*, vol. 56, no. 12, 2024, Art. no. 312.
- [30] F. Bodria, F. Giannotti, R. Guidotti, F. Naretto, D. Pedreschi, and S. Rinzivillo, "Benchmarking and survey of explanation methods for black box models," *Data Mining Knowl. Discov.*, vol. 37, no. 5, pp. 1719–1778, 2023.
- [31] R. Dwivedi et al., "Explainable AI (XAI): Core ideas, techniques, and solutions," *ACM Comput. Surv.*, vol. 55, no. 9, 2023, Art. no. 194.
- [32] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018.
- [33] R. Guidotti, "Counterfactual explanations and how to find them: Literature review and benchmarking," *Data Mining Knowl. Discov.*, vol. 38, no. 5, pp. 2770–2824, 2024.
- [34] K. Pearson, "LIII. On lines and planes of closest fit to systems of points in space," *London Edinburgh Dublin Philos. Mag. J. Sci.*, "LIII. On lines and planes of closest fit to systems of points in space," *London Edinburgh Dublin Philos. Mag. J. Sci.*, vol. 2, no. 11, pp. 559–572, 1901.
- [35] C. Eckart and G. Young, "The approximation of one matrix by another of lower rank," *Psychometrika*, vol. 1, no. 3, pp. 211–218, 1936.
- [36] D. Lee and H. S. Seung, "Algorithms for non-negative matrix factorization," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2000, pp. 535–541.
- [37] P. Comon, "Independent component analysis, a new concept?," *Signal Process.*, vol. 36, no. 3, pp. 287–314, 1994.
- [38] D. J. Bartholomew, "Spearman and the origin and development of factor analysis," *Brit. J. Math. Stat. Psychol.*, vol. 48, no. 2, pp. 211–220, 1995.
- [39] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.
- [40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [41] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, and D. Pedreschi, "A survey of methods for explaining black box models," *ACM Comput. Surv.*, vol. 51, no. 5, 2018, Art. no. 93.
- [42] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *J. Mach. Learn. Res.*, vol. 20, no. 177, pp. 1–81, 2019.
- [43] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [44] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 4765–4774.
- [45] S. Wachter, B. Mittelstadt, and C. Russell, "Counterfactual explanations without opening the black box: Automated decisions and the GDPR," *Harvard J. Law Technol.*, vol. 31, pp. 841–887, 2018.
- [46] S. Venkatasubramanian and M. Alfano, "The philosophical basis of algorithmic recourse," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2020, pp. 284–293.
- [47] A.-H. Karimi, G. Barthe, B. Sch lkopf, and I. Valera, "A survey of algorithmic recourse: Contrastive explanations and consequential recommendations," *ACM Comput. Surv.*, vol. 55, no. 5, 2022, Art. no. 95.
- [48] B. Salimi, R. Pradhan, J. Zhu, and B. Glavic, "Interpretable data-based explanations for fairness debugging," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2022, pp. 247–261.
- [49] J. Brophy, "Exit through the training data: A look into instance-attribution explanations and efficient data deletion in machine learning," Dept. Comput. Sci., Univ. Oregon, Eugene, OR, USA, 2020.
- [50] A.-H. Karimi, B. Sch lkopf, and I. Valera, "Algorithmic recourse: From counterfactual explanations to interventions," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2021, pp. 353–362.
- [51] R. Guidotti, A. Monreale, F. Giannotti, D. Pedreschi, S. Ruggieri, and F. Turini, "Factual and counterfactual explanations for black box decision making," *IEEE Intell. Syst.*, vol. 34, no. 6, pp. 14–23, Nov./Dec. 2019.
- [52] N. Tintarev and J. Masthoff, "A survey of explanations in recommender systems," in *Proc. IEEE Int. Conf. Data Eng. Workshops*, 2007, pp. 801–810.
- [53] Y. Zhang et al., "Explainable recommendation: A survey and new perspectives," *Found. Trends Inf. Retrieval*, vol. 14, no. 1, pp. 1–101, 2020.
- [54] C. Scheel, A. Castellanos, T. Lee, and E. W. De Luca, "The reason why: A survey of explanations for recommender systems," in *Proc. 10th Int. Workshop Adapt. Multimedia Retrieval*, 2014, pp. 67–84.
- [55] I. Nunes and D. Jannach, "A systematic review and taxonomy of explanations in decision support and recommender systems," *User Model. User-Adapted Interact.*, vol. 27, pp. 393–444, 2017.
- [56] A. Saldanha and H. Han, "A survey of explainable recommender systems," in *Proc. CS IT Int. Conf.*, 2024, pp. 159–171.
- [57] H. Yuan, H. Yu, S. Gui, and S. Ji, "Explainability in graph neural networks: A taxonomic survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5782–5799, May 2023.
- [58] M. Bugue o, R. Biswas, and G. de Melo, "Graph-based explainable AI: A comprehensive survey," *HAL*, Tech. Rep. hal-04660442, 2024.
- [59] Y. Li, J. Zhou, S. Verma, and F. Chen, "A survey of explainable graph neural networks: Taxonomy and evaluation metrics," 2022, *arXiv:2207.12599*.

- [60] A. Longa et al., "Explaining the explainers in graph neural networks: A comparative study," *ACM Comput. Surv.*, vol. 57, no. 5, 2024, Art. no. 120.
- [61] M. A. Prado-Romero, B. Prenkaj, G. Stilo, and F. Giannotti, "A survey on graph counterfactual explanations: Definitions, methods, evaluation, and research challenges," *ACM Comput. Surv.*, vol. 56, no. 7, 2024, Art. no. 171.
- [62] M. Nandan, S. Mitra, and D. De, "GraphXAI: A survey of graph neural networks (GNNs) for explainable AI (XAI)," *Neural Comput. Appl.*, vol. 37, no. 17, pp. 10949–11000, 2025.
- [63] E. Rajabi and K. Etminani, "Knowledge-graph-based explainable AI: A systematic review," *J. Inf. Sci.*, vol. 50, no. 4, pp. 1019–1029, 2024.
- [64] F. Hohman, M. Kahng, R. Pienta, and D. H. Chau, "Visual analytics in deep learning: An interrogative survey for the next frontiers," *IEEE Trans. Vis. Comput. Graph.*, vol. 25, no. 8, pp. 2674–2693, Aug. 2019.
- [65] S. Mohseni, N. Zarei, and E. D. Ragan, "A multidisciplinary survey and framework for design and evaluation of explainable AI systems," *ACM Trans. Interactive Intell. Syst.*, vol. 11, no. 3/4, 2021, Art. no. 24.
- [66] R. Yu and L. Shi, "A user-based taxonomy for deep learning visualization," *Vis. Inform.*, vol. 2, no. 3, pp. 147–154, 2018.
- [67] R. Tomsett, D. Braines, D. Harborne, A. Preece, and S. Chakraborty, "Interpretable to whom? A role-based model for analyzing interpretable machine learning systems," in *Proc. ICML Workshop Hum. Interpretability Mach. Learn.*, 2018.
- [68] U. Bhatt et al., "Explainable machine learning in deployment," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2020, pp. 648–657.
- [69] S. R. Hong, J. Hullman, and E. Bertini, "Human factors in model interpretability: Industry practices, challenges, and needs," in *Proc. ACM Hum.-Comput. Interact.*, vol. 4, no. CSCW1, 2020, Art. no. 68.
- [70] H. Suresh, S. R. Gomez, K. K. Nam, and A. Satyanarayan, "Beyond expertise and roles: A framework to characterize the stakeholders of interpretable machine learning and their needs," in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, 2021, Art. no. 74.
- [71] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 1135–1144.
- [72] H. van der Linden, H. Haned, and E. Kanoulas, "Global aggregations of local explanations for black box models," 2019, *arXiv:1907.03039*.
- [73] V. Papanikou, D. Pla Karidi, E. Pitoura, E. Panagiotou, and E. Ntoutsi, "Explanations as bias detectors: A critical study of local post-hoc XAI methods for fairness exploration," 2025, *arXiv:2505.00802*.
- [74] V. Papanikou, D. Pla Karidi, E. Pitoura, E. Panagiotou, and E. Ntoutsi, "MIMOSA: A tool for fairness exploration through explanations," in *Proc. Int. Conf. Extending Database Technol.*, 2026, pp. 732–735.
- [75] V. Bhargava, M. Couceiro, and A. Napoli, "LimeOut: An ensemble approach to improve process fairness," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2020, pp. 475–491.
- [76] G. Alves, V. Bhargava, M. Couceiro, and A. Napoli, "Making ML models fairer through explanations: The case of limeout," in *Proc. Int. Conf. Anal. Images Social Netw. Texts*, 2021, pp. 3–18.
- [77] L. S. Shapley et al., "A value for n-person games," in *Contributions to the Theory of Games II*, H. W. Kuhn and A. W. Tucker, Eds. Princeton, NJ, USA: Princeton Univ. Press, 1953, pp. 307–317.
- [78] M. Sundararajan and A. Najmi, "The many Shapley values for model explanation," in *Proc. Int. Conf. Mach. Learn.*, 2020, Art. no. 859.
- [79] A. Jain, M. Ravula, and J. Ghosh, "Biased models have biased explanations," 2020, *arXiv:2012.10986*.
- [80] Z. Wang, C. Huang, K. Tang, and X. Yao, "Procedural fairness in machine learning," *J. Artif. Intell. Res.*, vol. 85, 2026, Art. no. 20.
- [81] J. Cesaro and F. G. Cozman, "Measuring unfairness through game-theoretic interpretability," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2020, pp. 253–264.
- [82] P. A. Grabowicz, N. Perello, and A. Mishra, "Marrying fairness and explainability in supervised learning," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2022, pp. 1905–1916.
- [83] T. Begley, T. Schwedes, C. Frye, and I. Feige, "Explainability for fair machine learning," 2020, *arXiv:2010.07389*.
- [84] W. Pan, S. Cui, J. Bian, C. Zhang, and F. Wang, "Explaining algorithmic fairness through fairness-aware causal path decomposition," in *Proc. ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 1287–1297.
- [85] J. Zhu, R. Pradhan, B. Glavic, and B. Salimi, "Generating interpretable data-based explanations for fairness debugging using Gopher," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2022, pp. 2433–2436.
- [86] T. Surve and R. Pradhan, "Explaining fairness violations using machine unlearning," in *Proc. Int. Conf. Extending Database Technol.*, 2025, pp. 623–635.
- [87] S. Sharma, J. Henderson, and J. Ghosh, "CERTIFAI: A common framework to provide explanations and analyse the fairness and robustness of black-box models," in *Proc. AAAI/ACM Conf. AI Ethics Soc.*, 2020, pp. 166–172.
- [88] A. Karra, G. Vardakas, E. Pitoura, and A. Likas, "How much does cluster fairness cost? A counterfactual-based approach," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2026, pp. 311–326.
- [89] A. Kuratomi, E. Pitoura, P. Papapetrou, T. Lindgren, and P. Tsaparas, "Measuring the burden of (un) fairness using counterfactuals," in *Proc. Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2023, pp. 402–417.
- [90] S. Goethals, D. Martens, and T. Calders, "PreCoF: Counterfactual explanations for fairness," *Mach. Learn.*, vol. 113, no. 5, pp. 3111–3142, 2023.
- [91] B. Ustun, A. Spangher, and Y. Liu, "Actionable recourse in linear classification," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2019, pp. 10–19.
- [92] S. Galhotra, R. Pradhan, and B. Salimi, "Explaining black-box algorithms using probabilistic contrastive counterfactuals," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2021, pp. 577–590.
- [93] K. Rawal and H. Lakkaraju, "Beyond individualized recourse: Interpretable and interactive summaries of actionable recourses," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, Art. no. 1022.
- [94] D. Ley, S. Mishra, and D. Magazzeni, "GLOBE-CE: A translation-based approach for global counterfactual explanations," in *Proc. Int. Conf. Mach. Learn.*, 2023, Art. no. 797.
- [95] K. Kanamori, T. Takagi, K. Kobayashi, and Y. Ike, "Counterfactual explanation trees: Transparent and consistent actionable recourse with decision trees," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2022, pp. 1846–1870.
- [96] L. Kavouras et al., "Fairness aware counterfactuals for subgroups," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2023, Art. no. 2538.
- [97] C. Fragkathoulas, V. Papanikou, E. Pitoura, and E. Terzi, "FACEGroup: Feasible and actionable counterfactual explanations for group fairness," in *Proc. Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2025, pp. 41–59.
- [98] Y. Dong, S. Wang, Y. Wang, T. Derr, and J. Li, "On structural explanation of bias in graph neural networks," in *Proc. ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2022, pp. 316–326.
- [99] Z. Fu et al., "Fairness-aware explainable recommendation over knowledge graphs," in *Proc. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 69–78.
- [100] Y. Dong, S. Wang, J. Ma, N. Liu, and J. Li, "Interpreting unfairness in graph neural networks via training node attribution," in *Proc. AAAI Conf. Artif. Intell.*, 2023, Art. no. 836.
- [101] G. Medda, F. Fabbri, M. Marras, L. Boratto, and G. Fenu, "GNNUERS: Fairness explanation in GNNs for recommendation via counterfactual reasoning," *ACM Trans. Intell. Syst. Technol.*, vol. 16, no. 1, 2024, Art. no. 6.
- [102] L. Boratto, F. Fabbri, G. Fenu, M. Marras, and G. Medda, "Counterfactual graph augmentation for consumer unfairness mitigation in recommender systems," in *Proc. ACM Int. Conf. Inf. Knowl. Manage.*, 2023, pp. 3753–3757.
- [103] Y. Ma and L. Guo, "A counterfactual inspired framework for quantifying edge effects on GNNs fairness," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2024, pp. 7600–7604.
- [104] Z. Wang, Q. Zeng, W. Lin, M. Jiang, and K. C. Tan, "Generating diagnostic and actionable explanations for fair graph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2024, Art. no. 2420.
- [105] V. Gupta, P. Nokhiz, C. D. Roy, and S. Venkatasubramanian, "Equalizing recourse across groups," 2019, *arXiv:1909.03166*.
- [106] J. Von K ügelgen, A.-H. Karimi, U. Bhatt, I. Valera, A. Weller, and B. Schölkopf, "On the fairness of causal algorithmic recourse," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 9584–9594.
- [107] Y. Ge et al., "Explainable fairness in recommendation," in *Proc. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2022, pp. 681–691.
- [108] X. Wang, Q. Li, D. Yu, Q. Li, and G. Xu, "Counterfactual explanation for fairness in recommendation," *ACM Trans. Inf. Syst.*, vol. 42, no. 4, 2024, Art. no. 106.
- [109] J. Li, Y. Ren, M. Sanderson, and K. Deng, "Explaining recommendation fairness from a user/item perspective," *ACM Trans. Inf. Syst.*, vol. 43, no. 1, 2024, Art. no. 17.
- [110] L. Zafeiriou, P. Tsaparas, and E. Pitoura, "Counterfactual explanations for recommendation bias," in *Proc. Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2023, pp. 178–193.
- [111] Y. Deldjoo, A. Bellogin, and T. Di Noia, "Explaining recommender systems fairness and accuracy through the lens of data characteristics," *Inf. Process. Manage.*, vol. 58, no. 5, 2021, Art. no. 102662.

- [112] J. Li, Y. Moskovitch, and H. Jagadish, "Detection of groups with biased representation in ranking," in *Proc. IEEE Int. Conf. Data Eng.*, 2023, pp. 2167–2179.
- [113] Y. Moskovitch, J. Li, and H. Jagadish, "Dexer: Detecting and explaining biased representation in ranking," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2023, pp. 159–162.
- [114] A. N. Nikolakopoulos and G. Karypis, "RecWalk: Nearly uncoupled random walks for top-n recommendation," in *Proc. ACM Int. Conf. Web Search Data Mining*, 2019, pp. 150–158.
- [115] T. Wei, F. Feng, J. Chen, Z. Wu, J. Yi, and X. He, "Model-agnostic counterfactual reasoning for eliminating popularity bias in recommender system," in *Proc. ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 1791–1800.
- [116] Z. C. Lipton, A. Chouldechova, and J. McAuley, "Does mitigating ML's impact disparity require treatment disparity?," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 8136–8146.
- [117] J. Germino, Y. Zhao, T. Derr, N. Moniz, and N. V. Chawla, "Explanation difference: Bridging procedural and distributional fairness," in *Proc. AAAI/ACM Conf. AI Ethics Soc.*, 2025, pp. 1078–1090.
- [118] S. Krishna et al., "The disagreement problem in explainable machine learning: A practitioner's perspective," *Trans. Mach. Learn. Res.*, 2024.
- [119] P. Zschech, S. Weinzierl, and M. Kraus, "Inherently interpretable machine learning: A contrasting paradigm to post-hoc explainable AI," *Bus. Inf. Syst. Eng.*, vol. 68, no. 2, pp. 445–463, 2026.
- [120] U. Aivodji, H. Arai, O. Fortineau, S. Gambs, S. Hara, and A. Tapp, "Fairwashing: The risk of rationalization," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 161–170.
- [121] U. Aivodji, H. Arai, S. Gambs, and S. Hara, "Characterizing the risk of fairwashing," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2021, pp. 14822–14834.
- [122] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proc. AAAI/ACM Conf. AI Ethics Soc.*, 2020, pp. 180–186.
- [123] H. Hwang, A. Bell, J. Fonseca, V. Platsika, J. Stoyanovich, and S. E. Whang, "SHAP-based explanations are sensitive to feature representation," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2025, pp. 1588–1601.
- [124] J. Schoeffer, M. De-Arteaga, and N. Kuehl, "Explanations, fairness, and appropriate reliance in human-AI decision-making," in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, 2024, Art. no. 836.
- [125] A. Balagopalan, H. Zhang, K. Hamidieh, T. Hartvigsen, F. Rudzicz, and M. Ghassemi, "The road to explainability is paved with bias: Measuring the fairness of explanations," in *Proc. ACM Conf. Fairness Accountability Transparency*, 2022, pp. 1194–1206.
- [126] J. Dai, S. Upadhyay, U. Aivodji, S. H. Bach, and H. Lakkaraju, "Fairness via explanation quality: Evaluating disparities in the quality of post hoc explanations," in *Proc. AAAI/ACM Conf. AI Ethics Soc.*, 2022, pp. 203–214.
- [127] A. Artelt et al., "Evaluating robustness of counterfactual explanations," in *Proc. IEEE Symp. Ser. Comput. Intell.*, 2021, pp. 1–9.
- [128] A. Kuratomi et al., "CounterFair: Group counterfactuals for bias detection, mitigation and subgroup identification," in *Proc. IEEE Int. Conf. Data Mining*, 2024, pp. 181–190.
- [129] A. Kuratomi et al., "Subgroup fairness based on shared counterfactuals," *Knowl. Inf. Syst.*, vol. 67, no. 11, pp. 10863–10901, 2025.
- [130] C. Agarwal, O. Queen, H. Lakkaraju, and M. Zitnik, "Evaluating explainability for graph neural networks," *Sci. Data*, vol. 10, no. 1, 2023, Art. no. 144.
- [131] B. Wen, Y. Feng, Y. Zhang, and C. Shah, "Towards generating robust, fair, and emotion-aware explanations for recommender systems," 2022, *arXiv:2208.08017*.
- [132] D. Pla Karidi and E. Pitoura, "Path-based summary explanations for graph recommenders," in *Proc. IEEE Int. Conf. Data Eng.*, 2025, pp. 973–986.
- [133] C. Fragkathoulas, E. Psaroudaki, T. Palpanas, and E. Pitoura, "GALACTIC: Global and local agnostic counterfactuals for time-series clustering," 2026, *arXiv:2603.05318*.



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